



Electric Field and Potential Profile

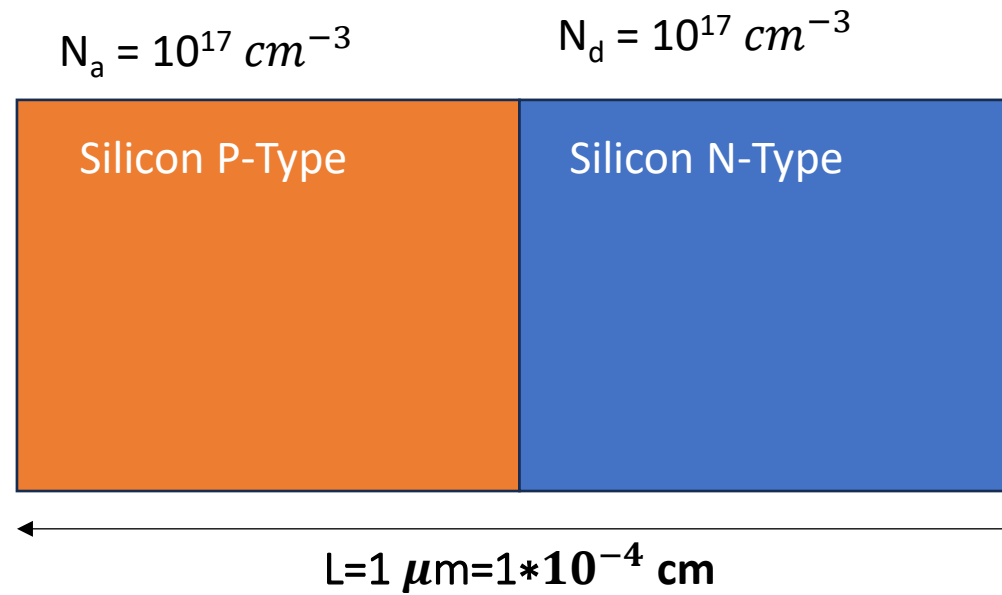
Faculty – Dr. Naresh Kumar Emani

Name- Bipul Biswas

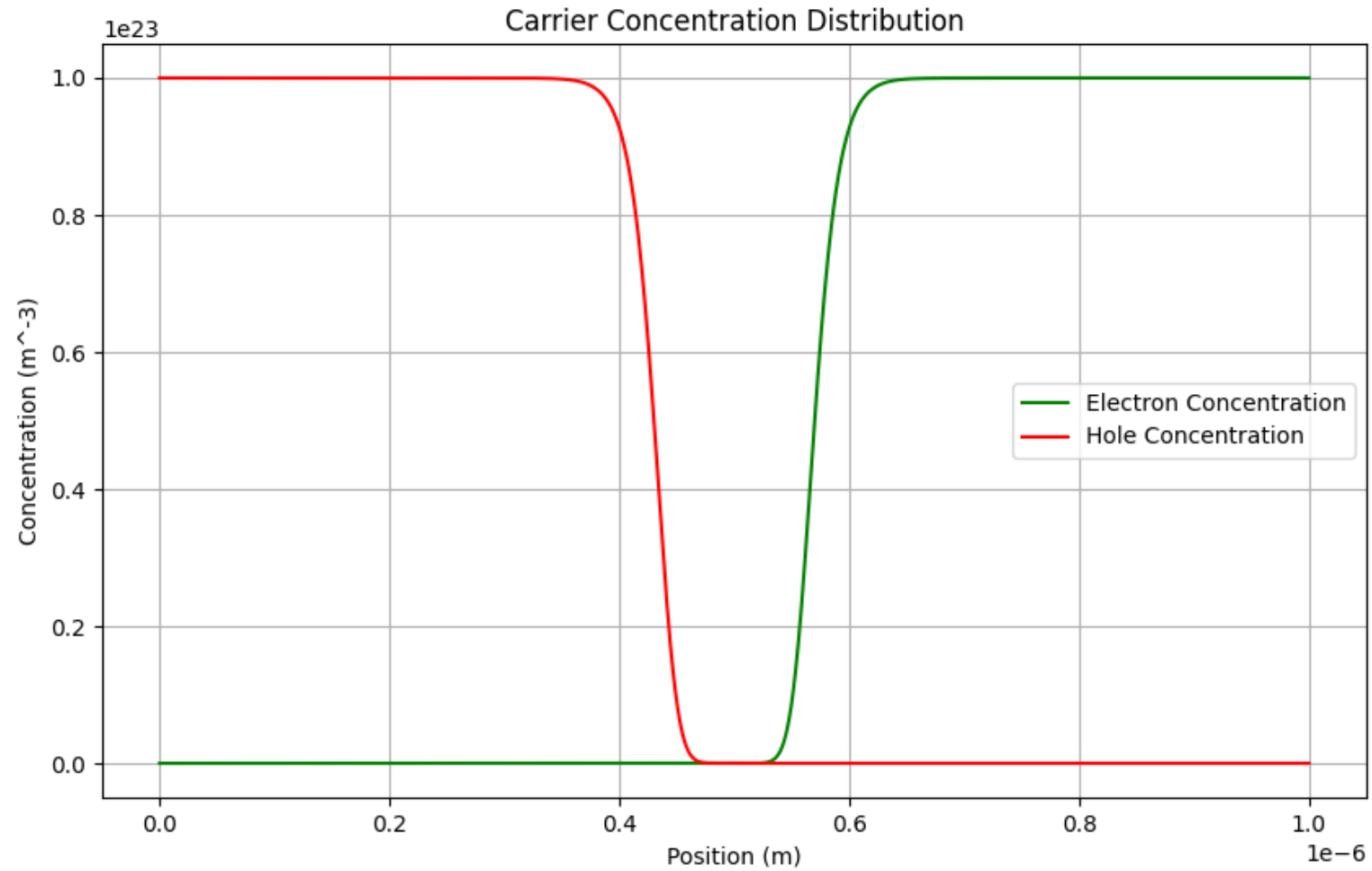
Roll No. EE24MTECH12003



Doping profile of material-1

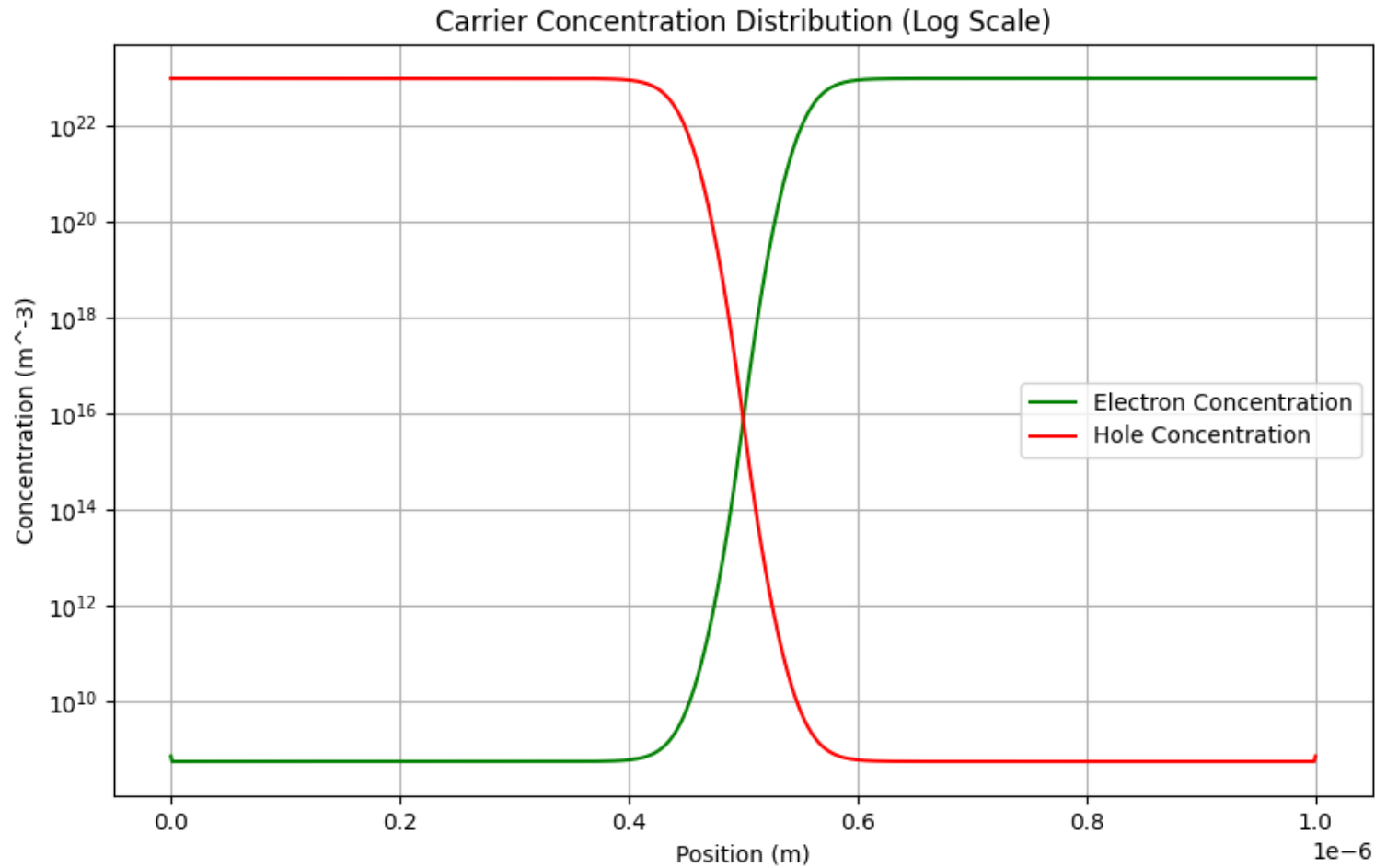


Electrons and Holes Profile

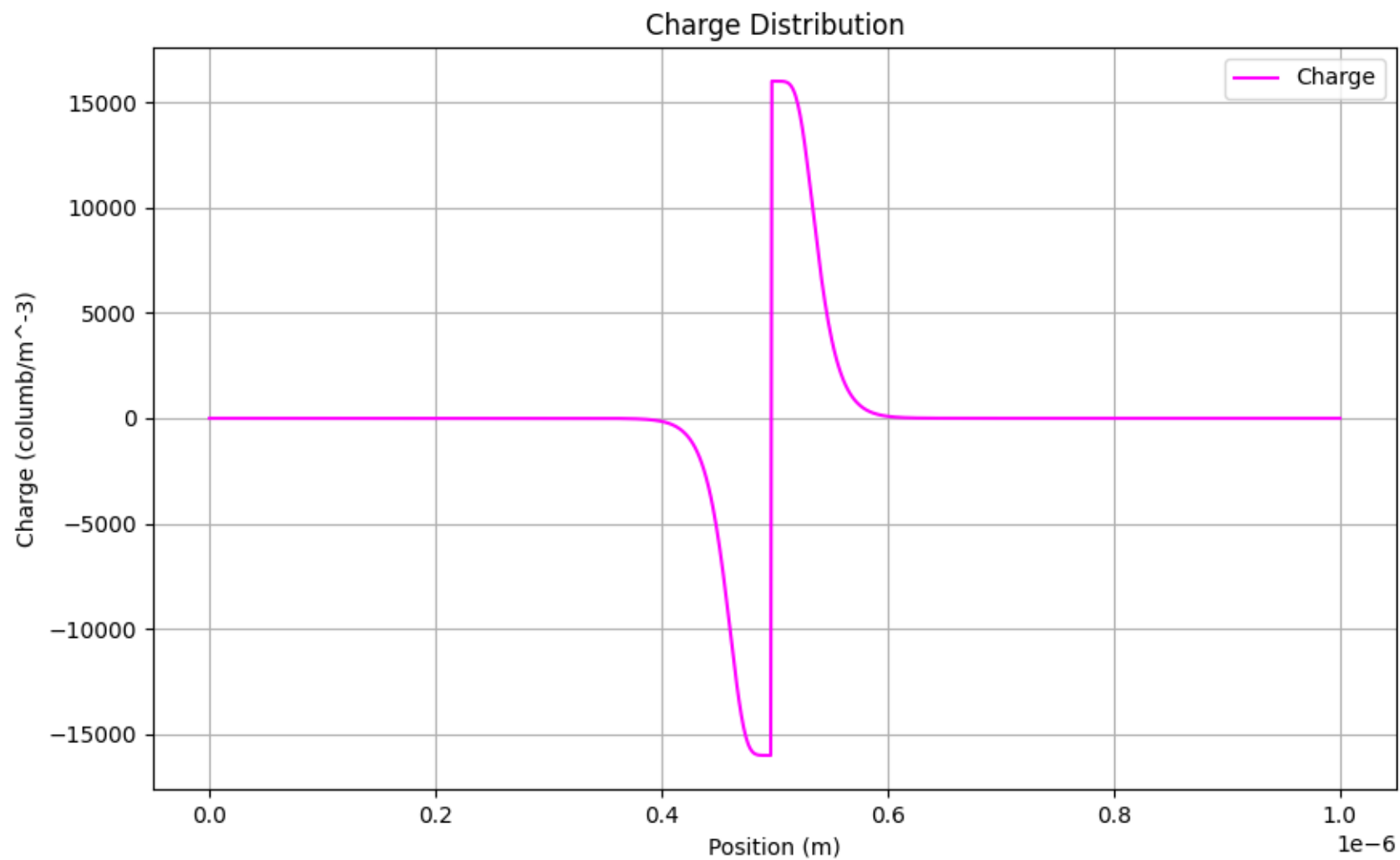




Electrons and Holes Profile

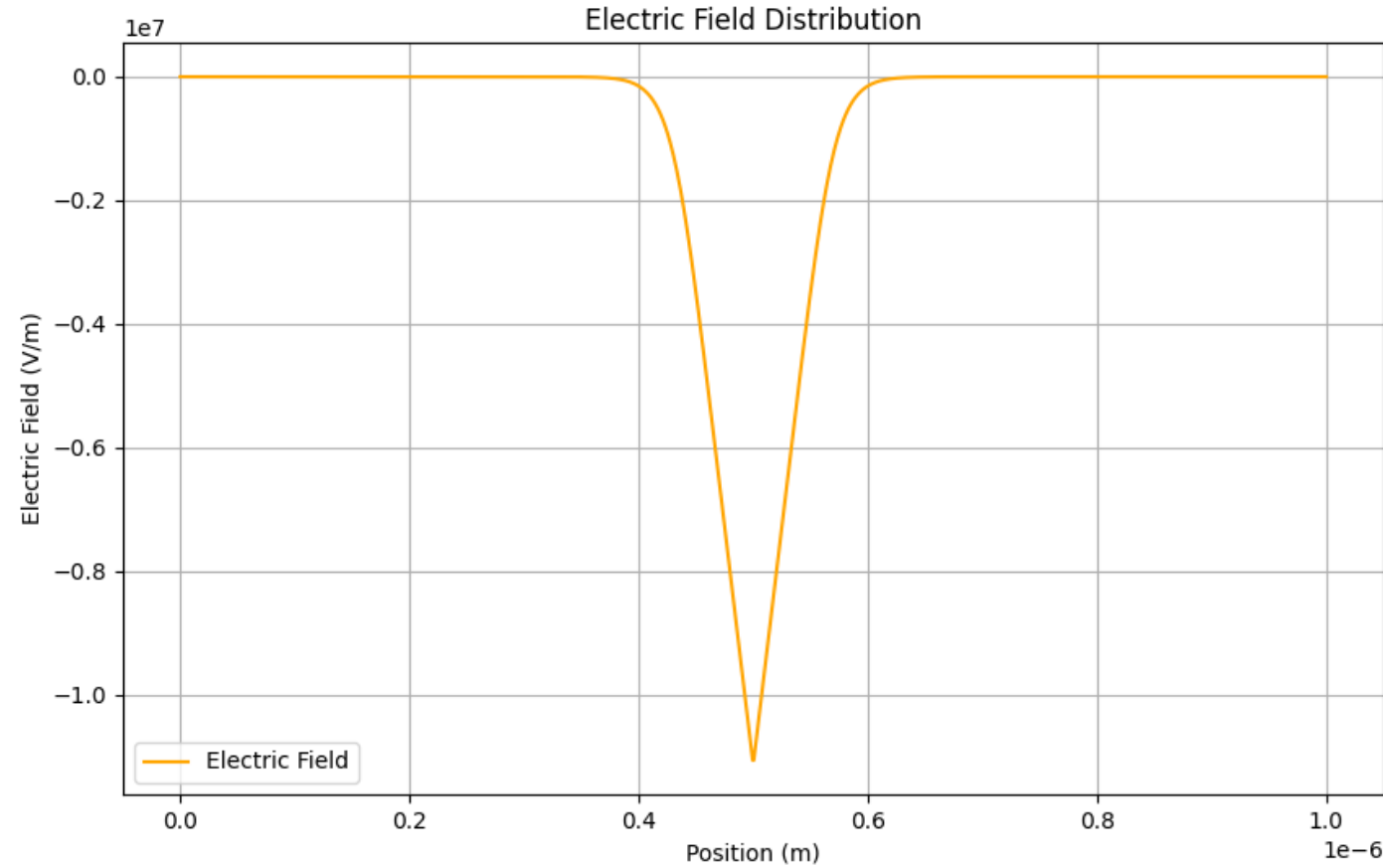


Charge Profile





Electric Field Profile



	Electric Field (V/m)
--	----------------------

Maximum	-1.10535e+07
---------	--------------

Minimum	-1.94289e-06
---------	--------------



$$W = \sqrt{\frac{2 \times \epsilon}{4} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) V_j}$$

$$W = \sqrt{\frac{2 \times 11.7 \times \epsilon_0}{1.6 \times 10^{-19}} \left(10^{17} + 10^{17} \right) 0.838}$$

$$W/2 = 7.36 \times 10^{-6} \text{ cm}$$

$$E = -13.8 - 46.8, 28 \times 10^7$$

$$|E| = -1.3546 \times 10^8 \text{ V/m}$$

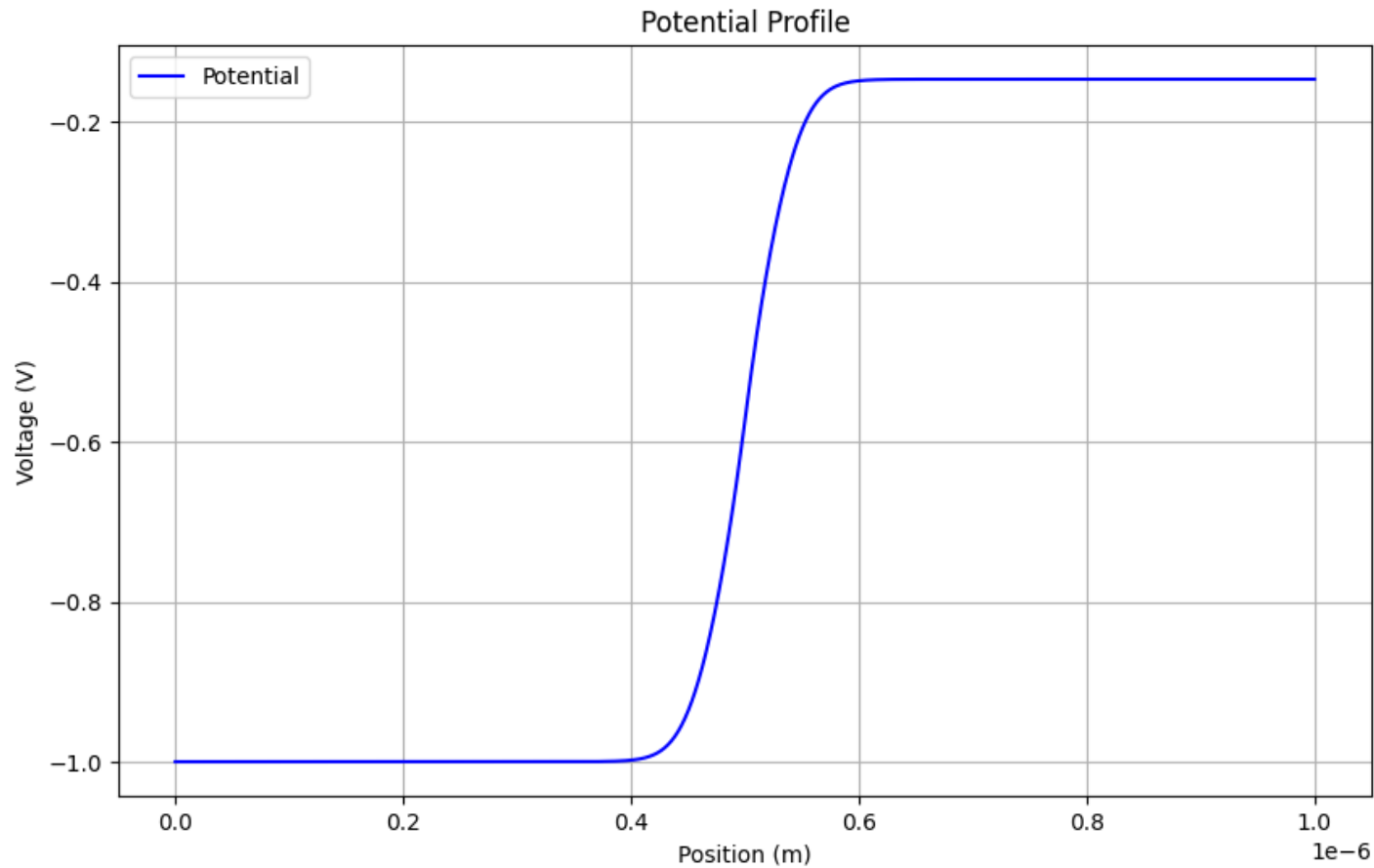
$$500^{\mu} \text{ Mode } E = -1.105 \times 10^8 \text{ V/m}$$

Shot on OnePlus

27 September 2024 at 13:56



Potential profile



Voltage (V)	
Maximum	-0.14650
minimum	-0.99924

Potential Difference = 0.8527 V



$$0.026 \ln \left(\frac{10^{46}}{10^{36.32}} \right)$$

$$= 0.838 \checkmark$$

μ
600 Node

μ
400 Node

$$= 0.99705 - (-0.148)$$

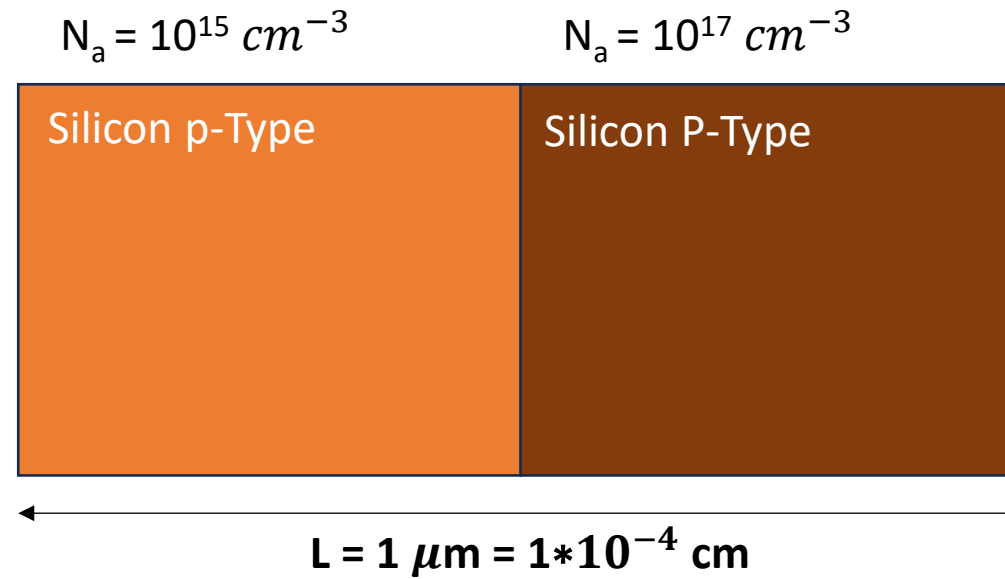
$$= 0.849$$

Shot on OnePlus

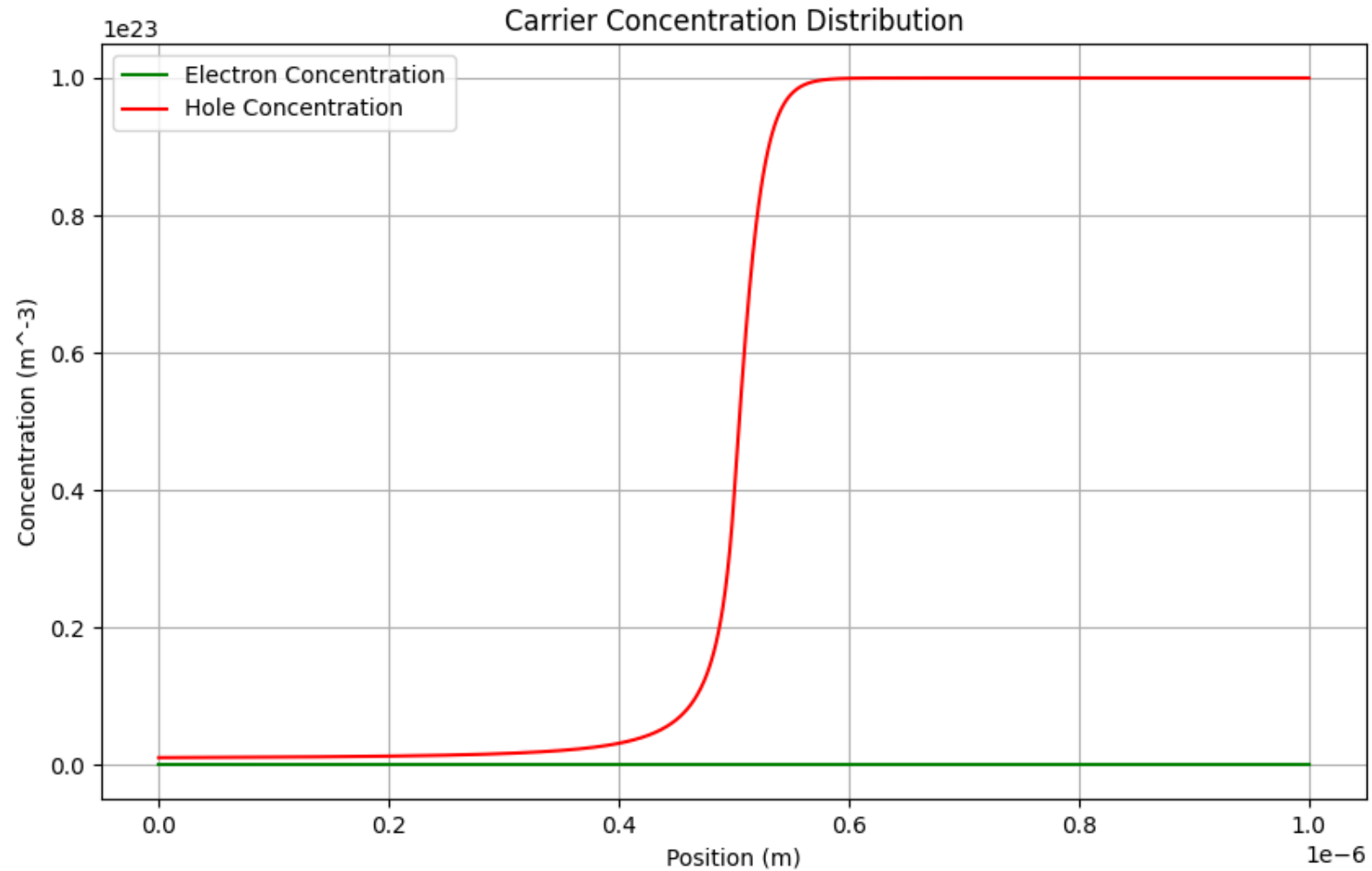
27 September 2024 at 13:55



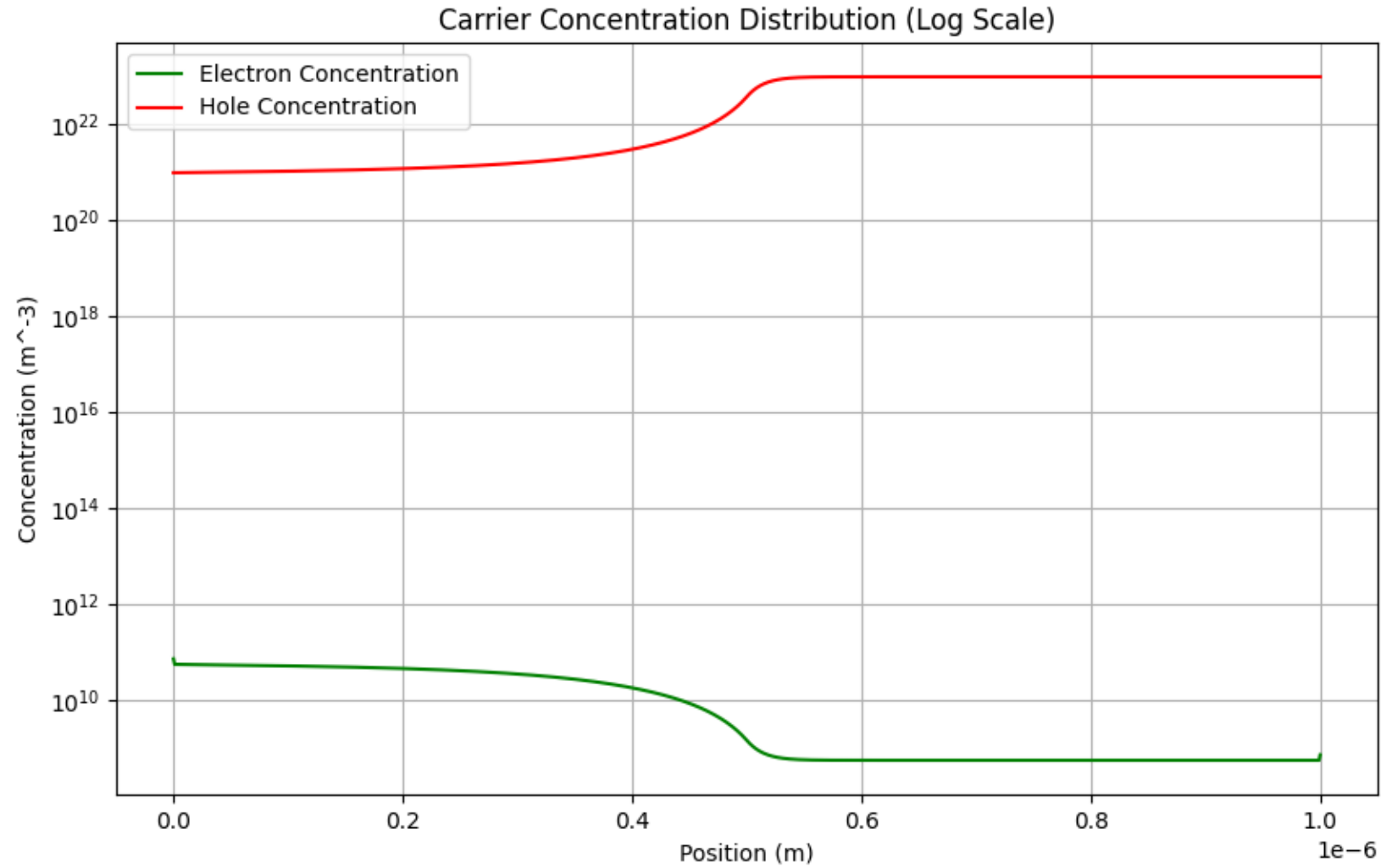
Doping profile of material-2



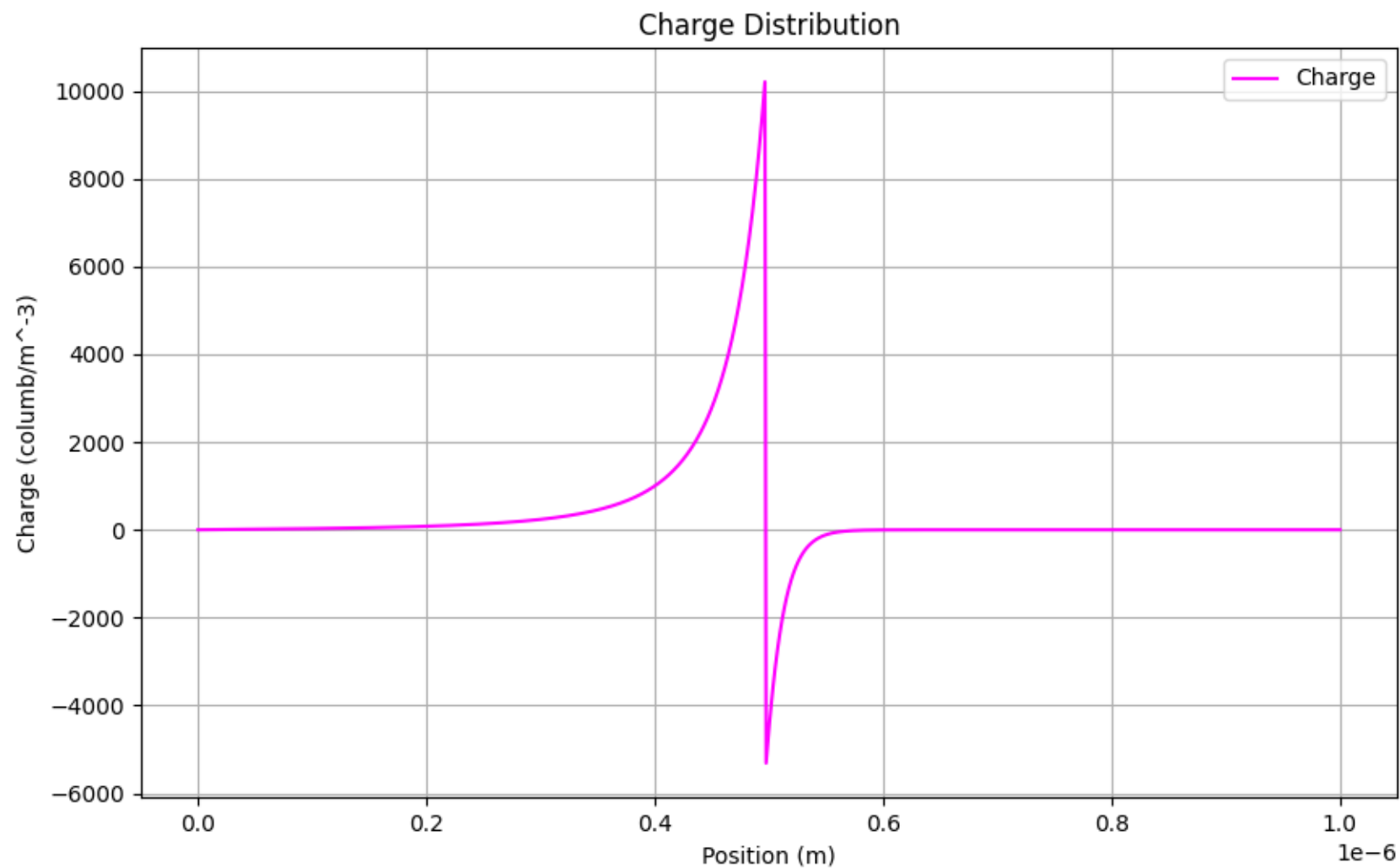
Electrons and holes Profile



Electrons and Holes Profile

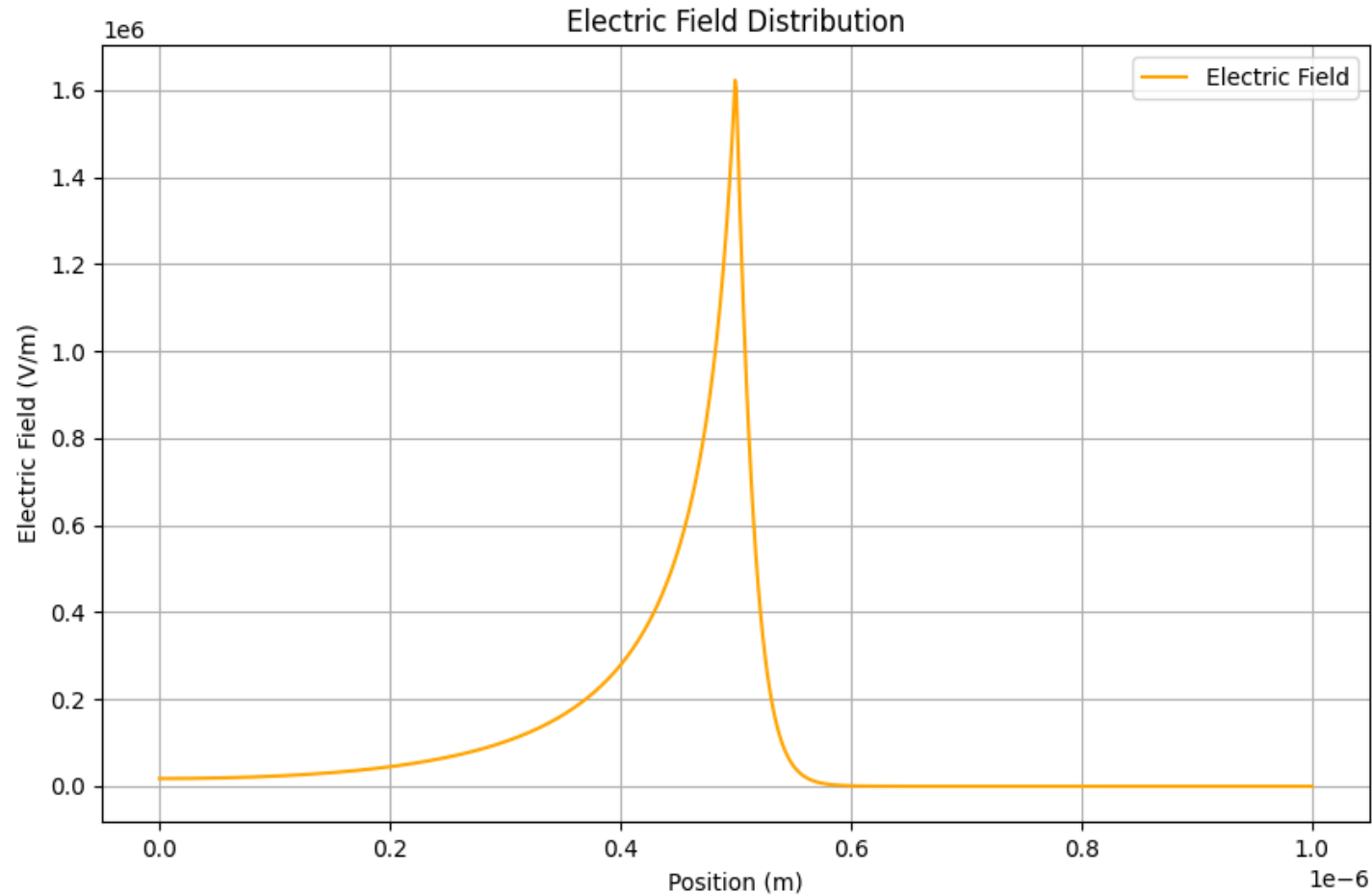


Charge Profile





Electric Field Profile

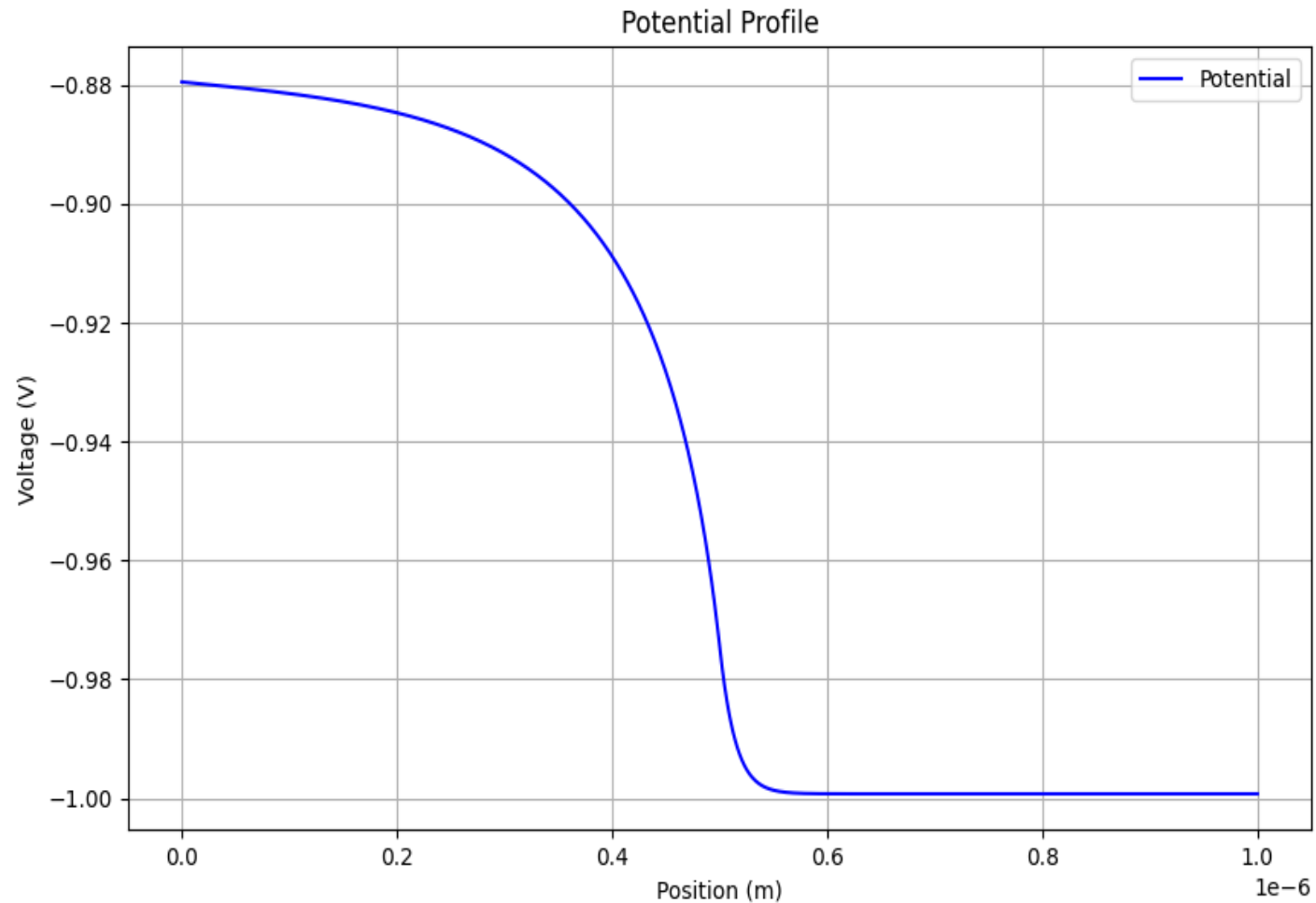


Electric Field (V/m)

Maximum	1.62277×10^6
Minimum	1.88738×10^{-6}



Potential Profile



	Voltage (V)
--	-------------

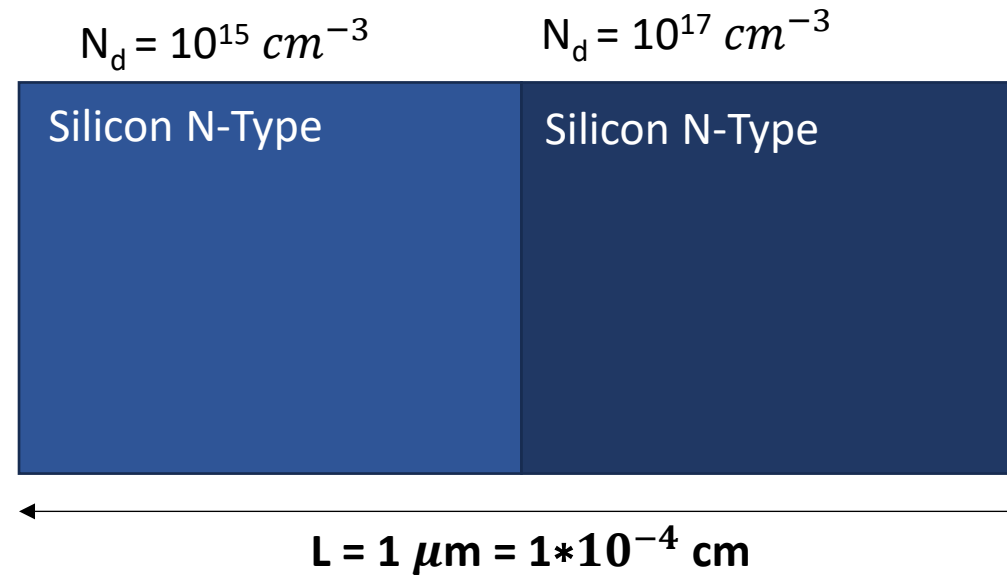
Maximum	-0.879511
---------	-----------

minimum	-0.999245
---------	-----------

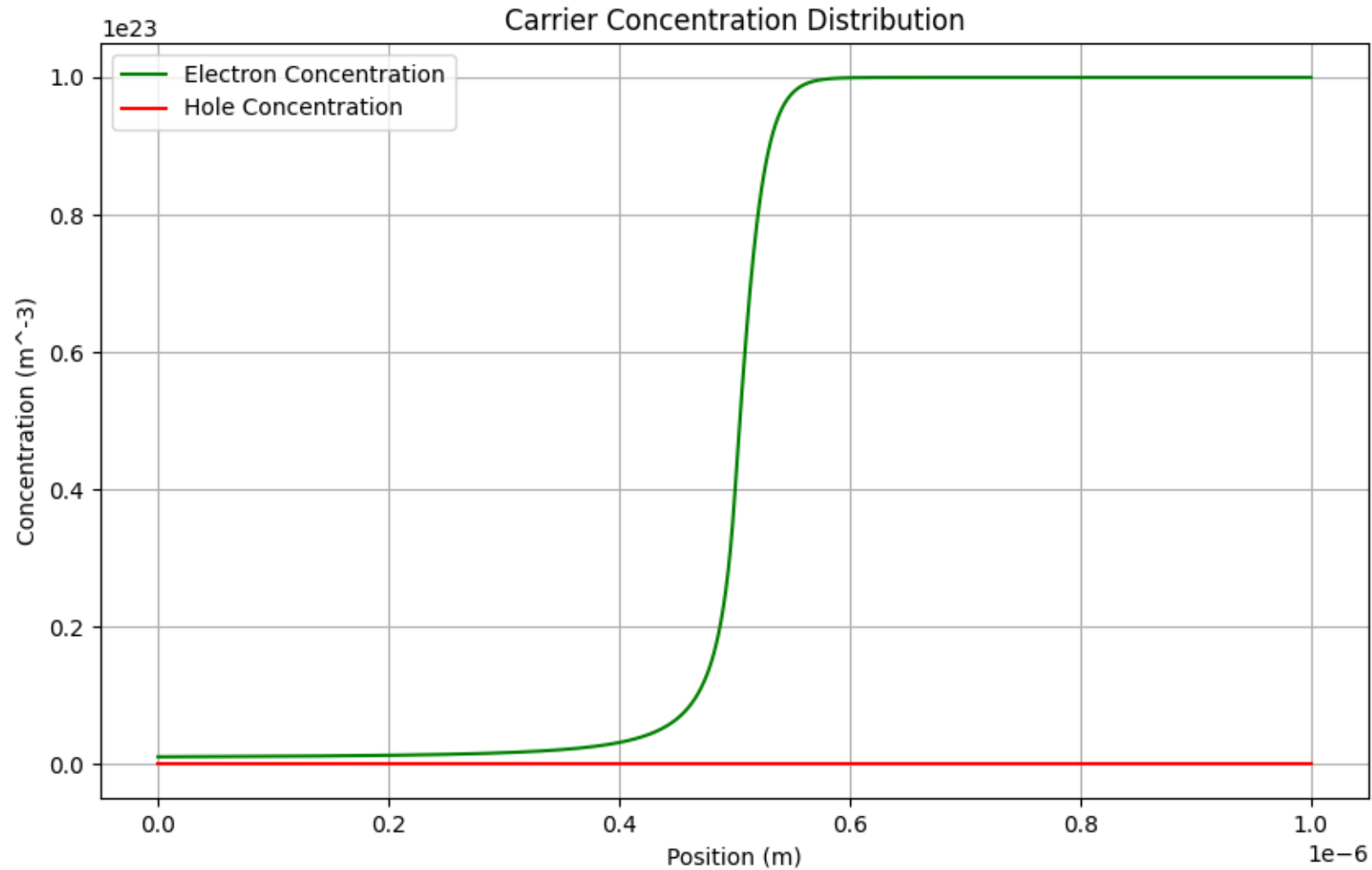
Potential Difference = 0.1197 V



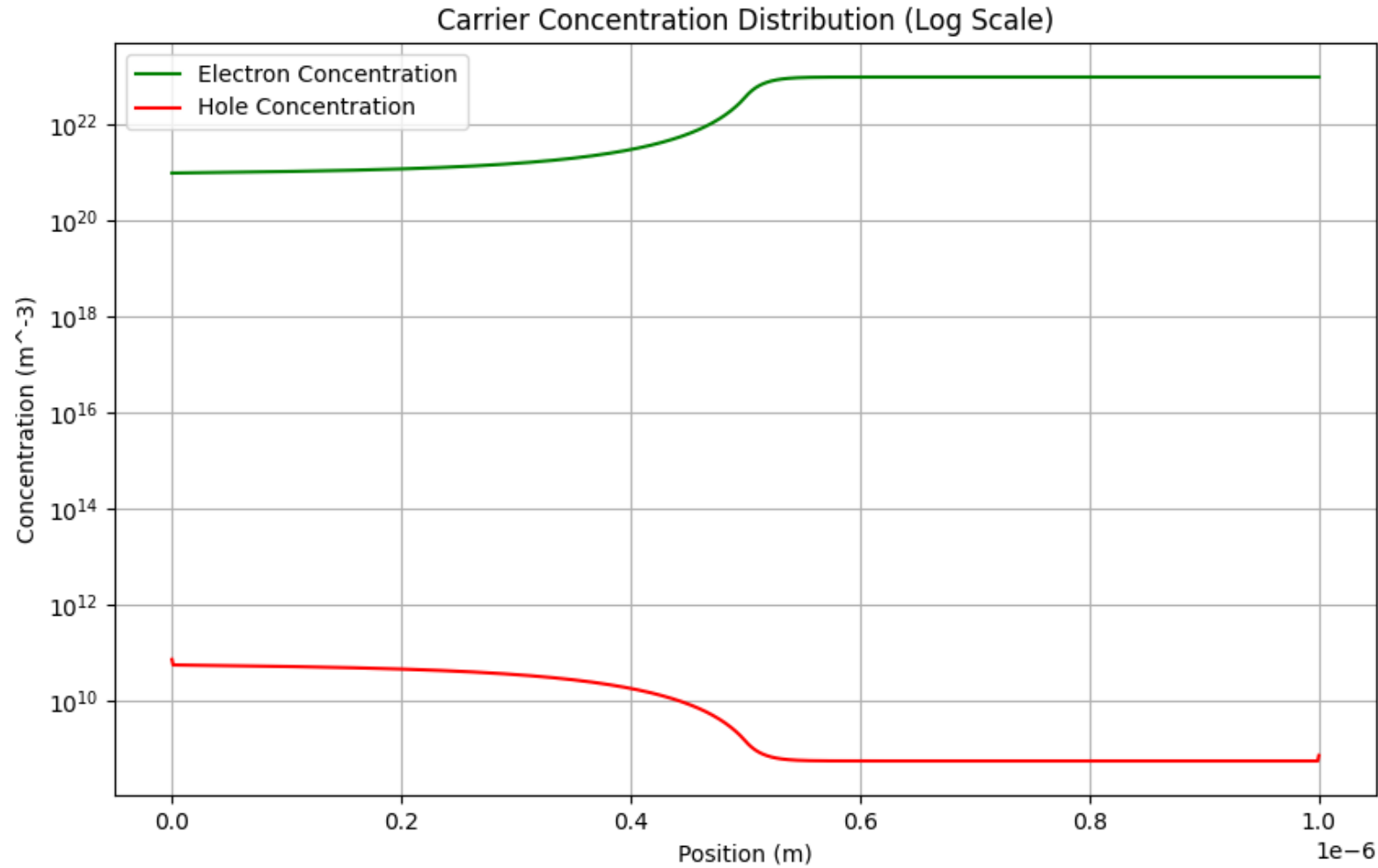
Doping profile of material-3



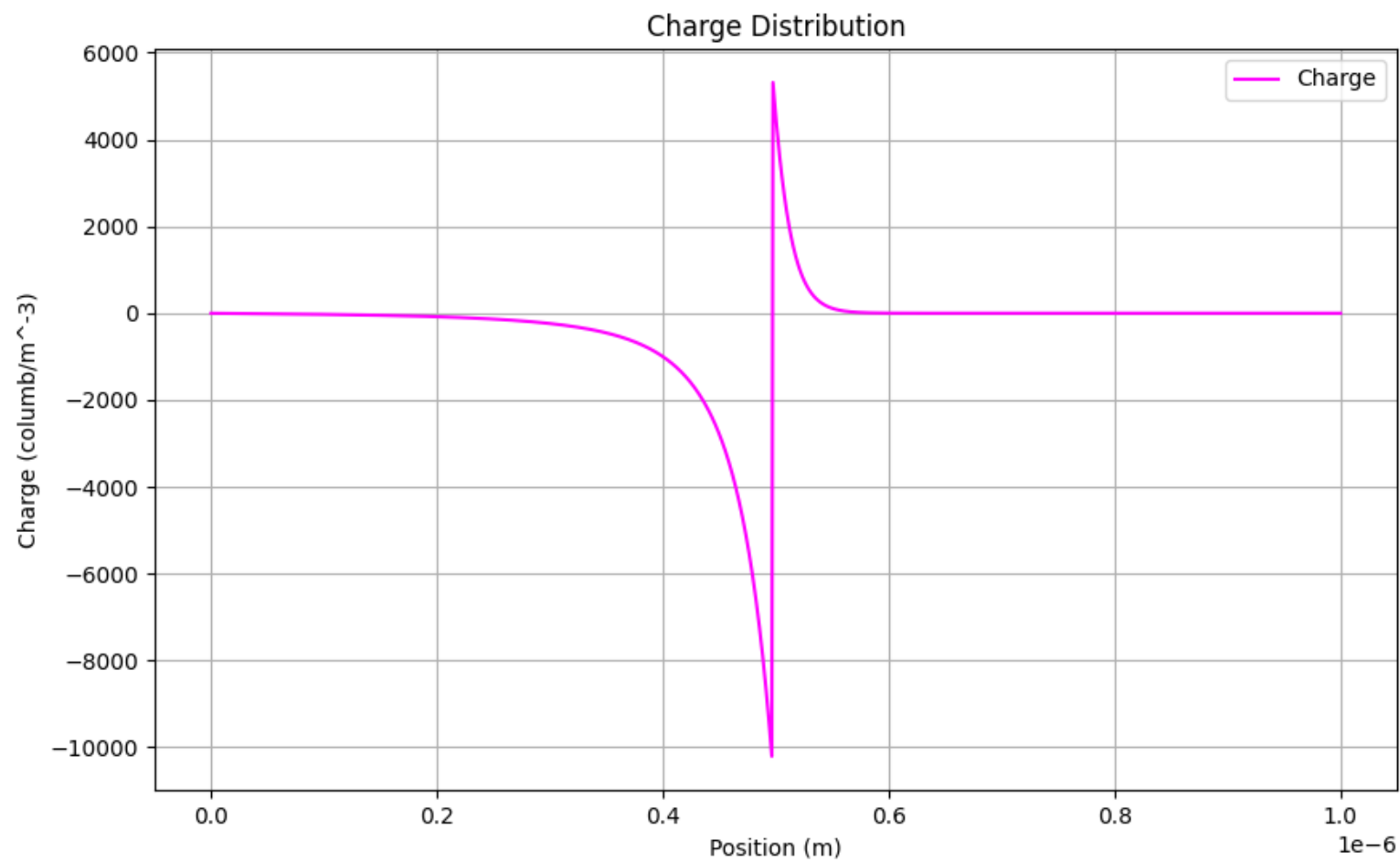
Electrons and Holes Profile



Electrons and Holes Profile

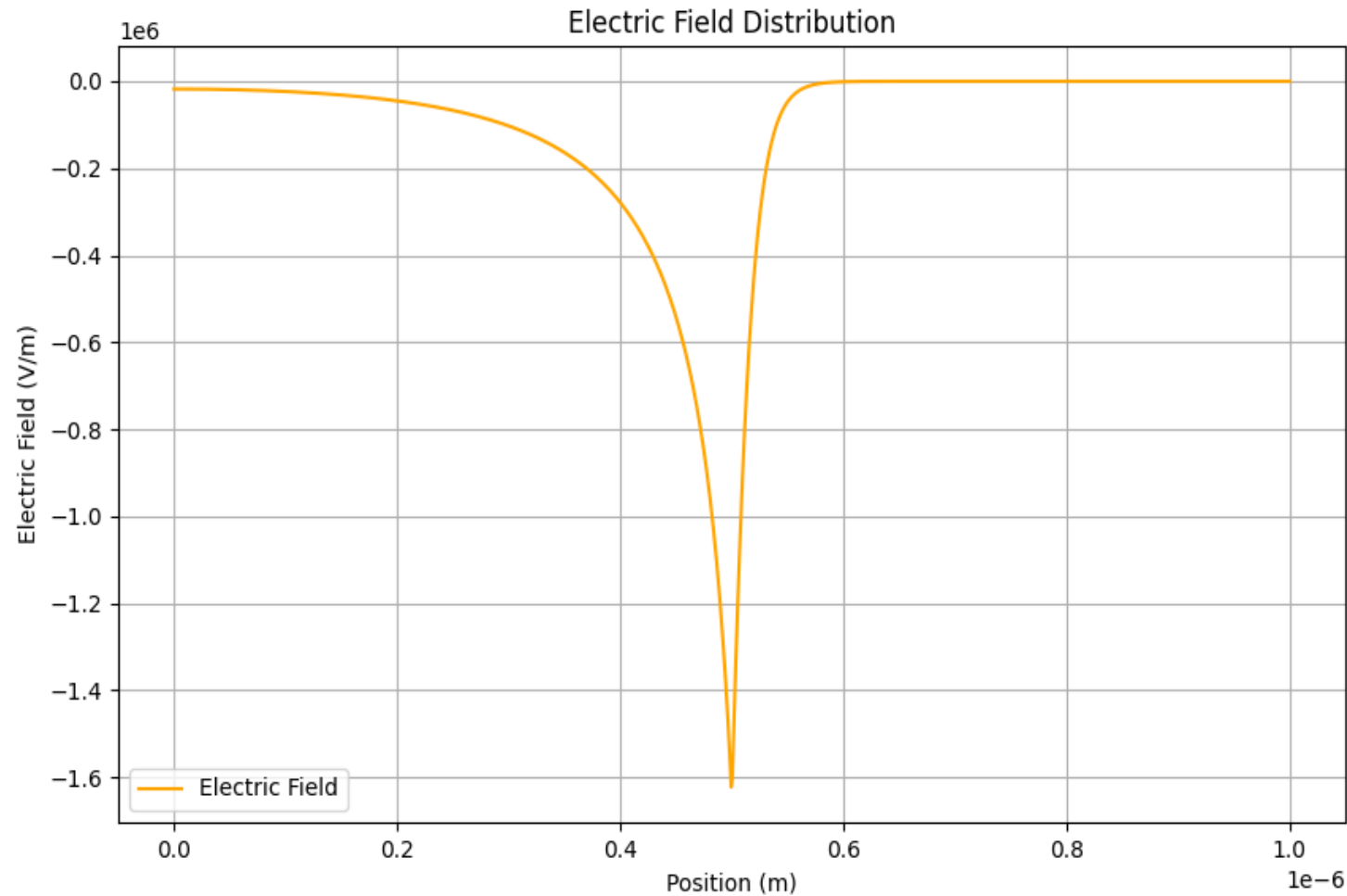


Charge Profile





Electric Field Profile



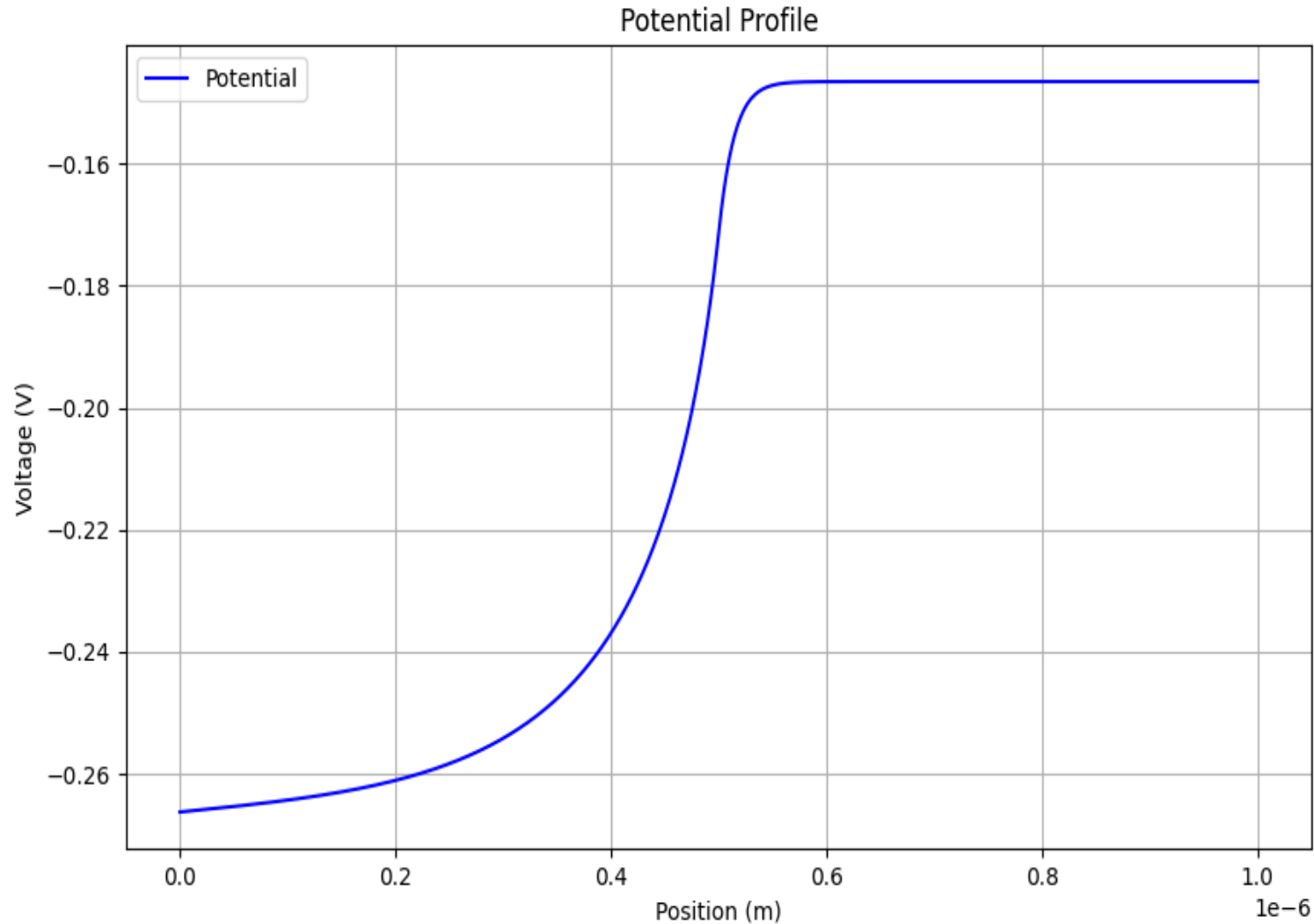
Electric Field (V/m)

Maximum	-1.62277e+06
---------	--------------

Minimum	-1.91513e-06
---------	--------------



Potential Profile

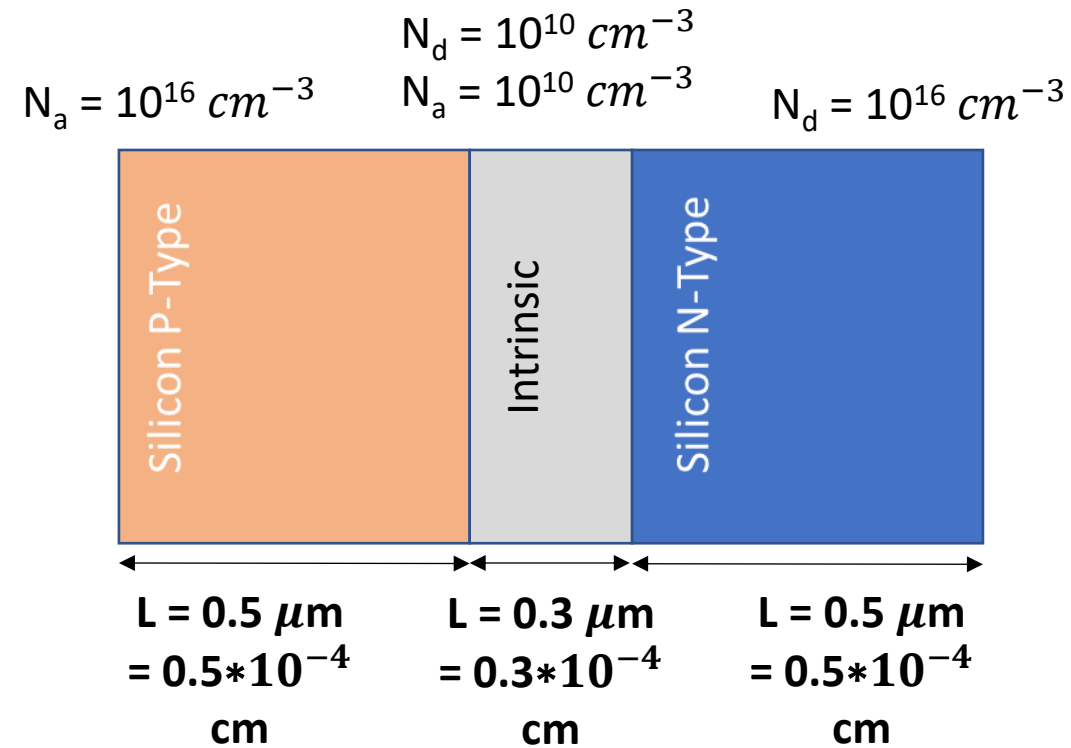


Voltage (V)	
Maximum	-0.14650
minimum	-0.26623

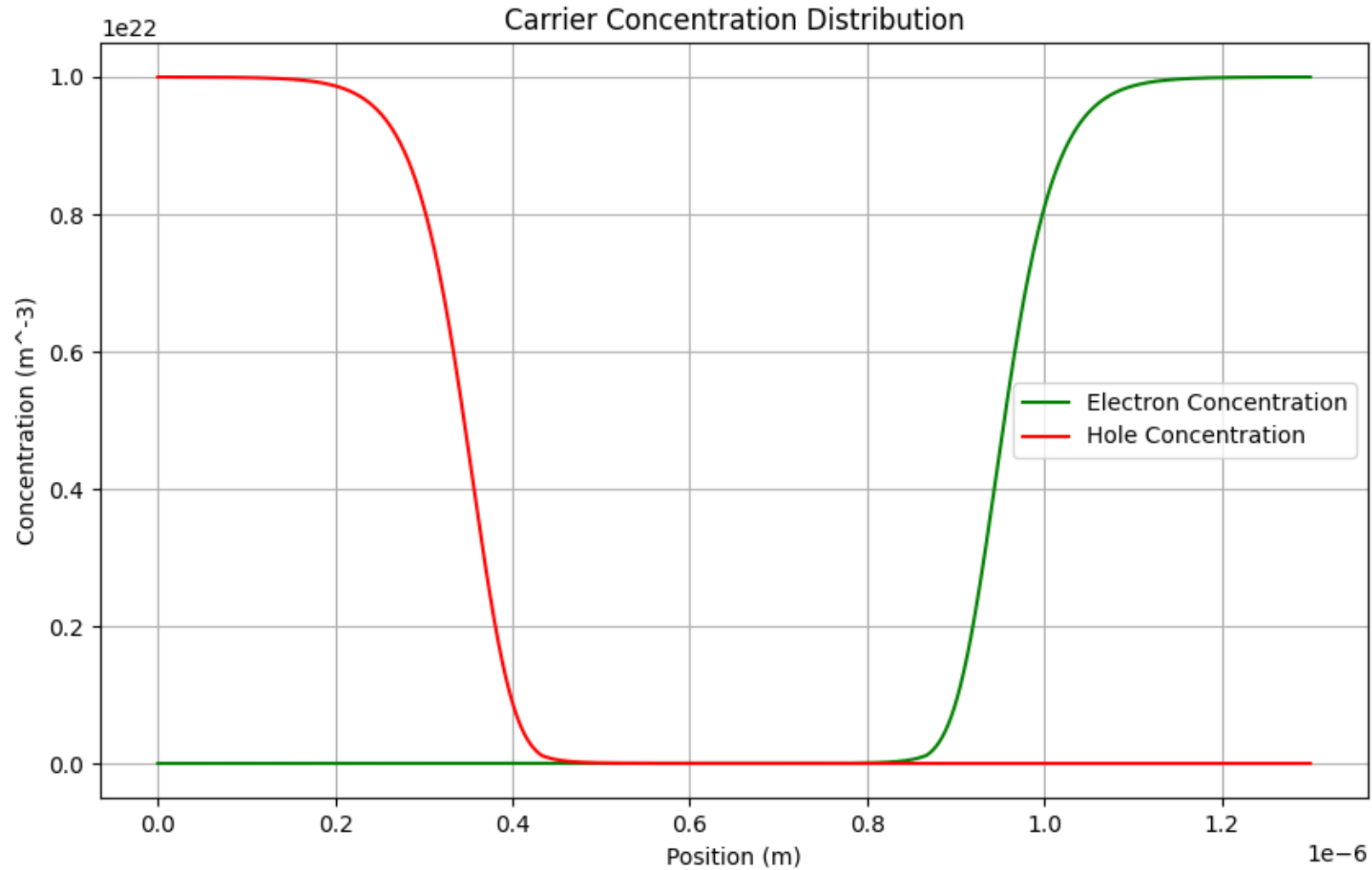
Potential Difference = -0.11973 V



Doping profile of material-4

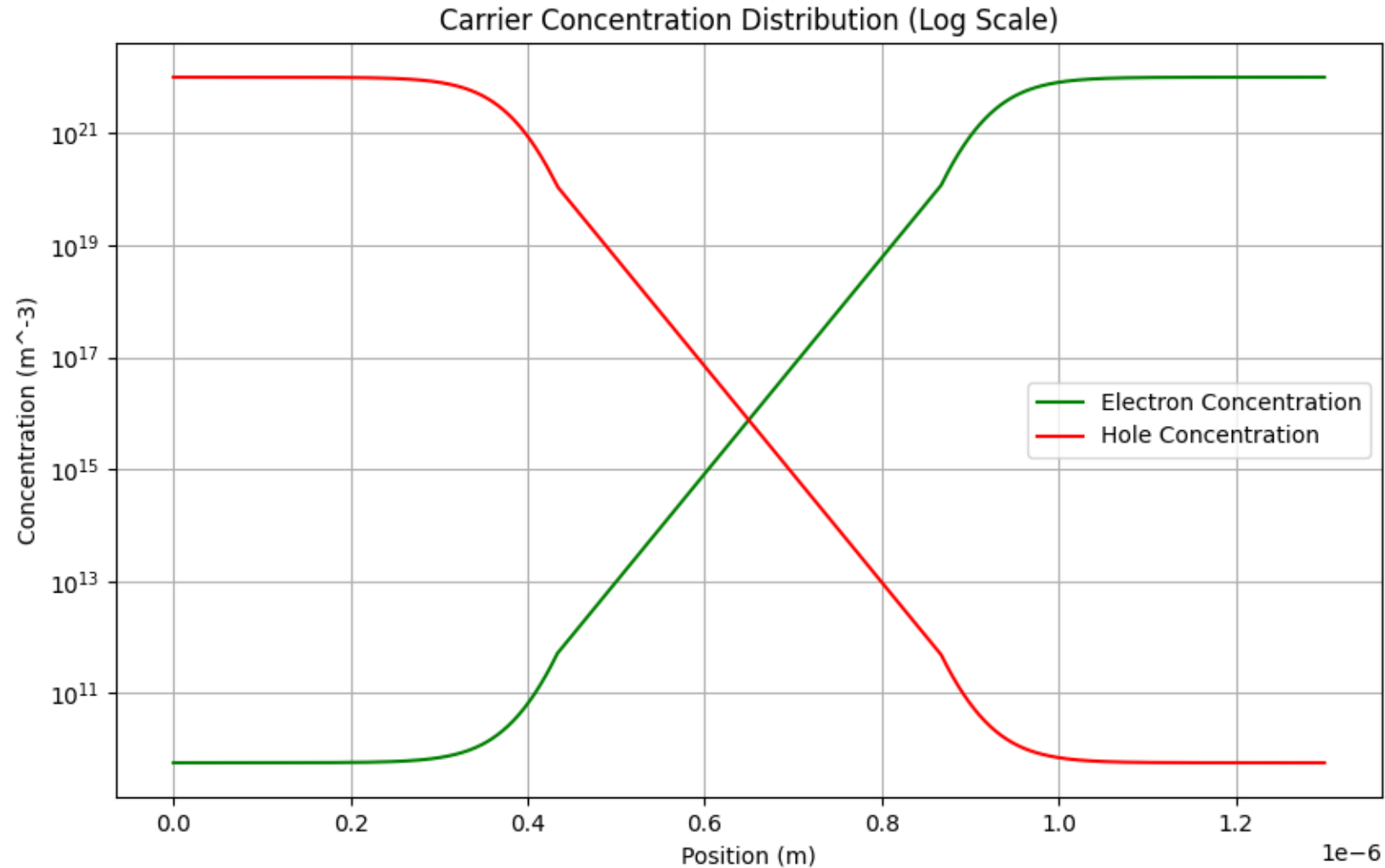


Electrons and holes Profile

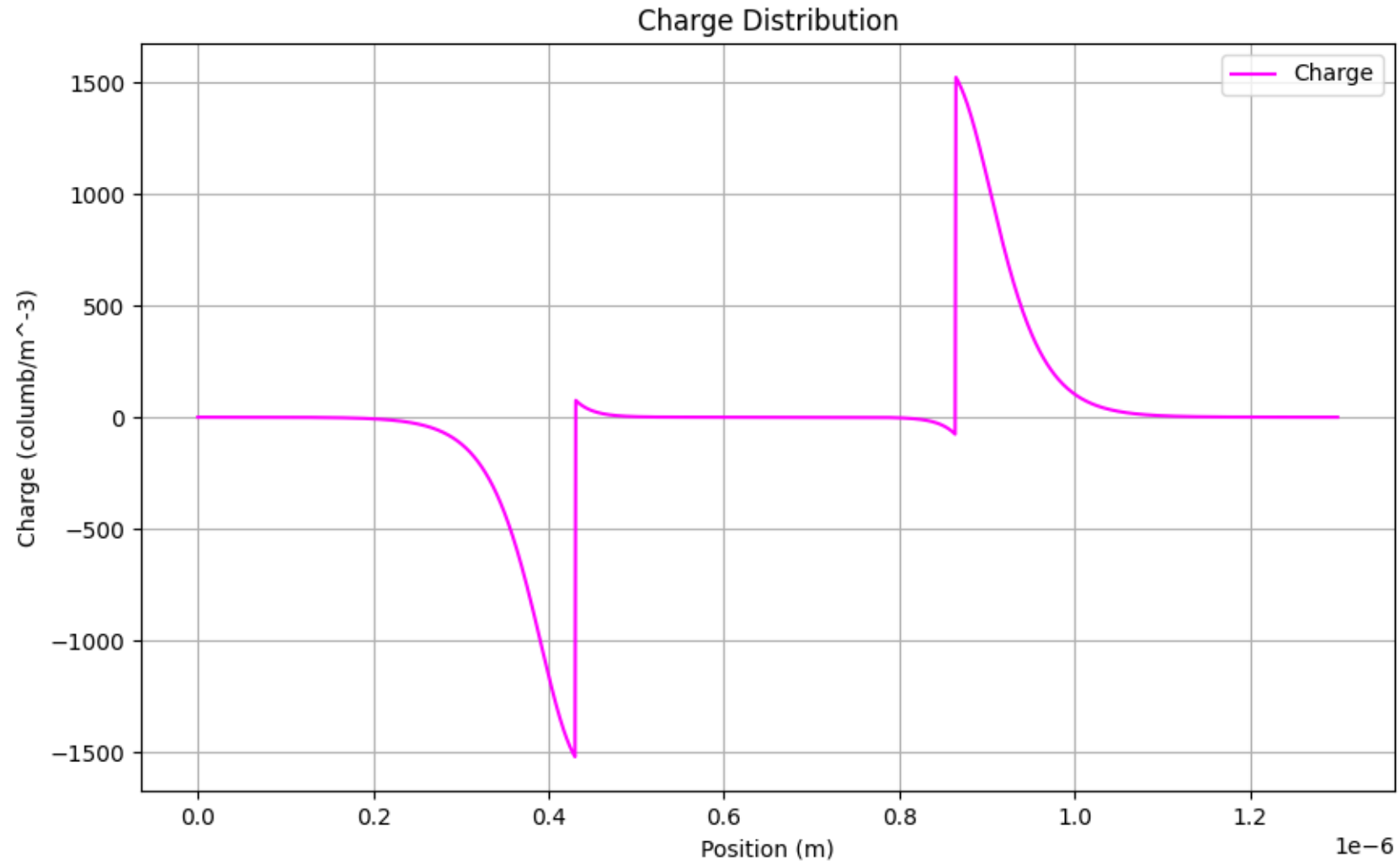




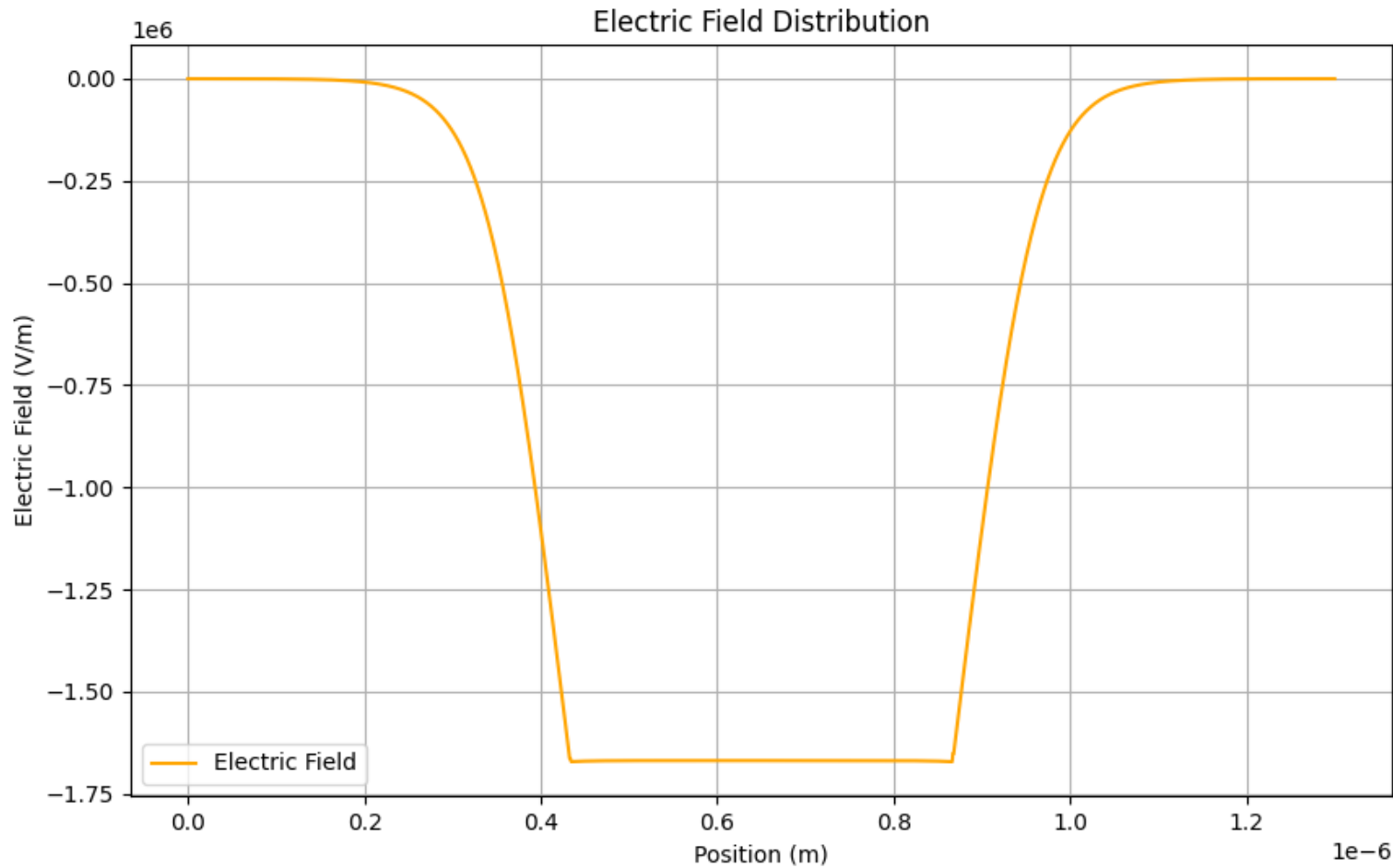
Electrons and Holes Profile



Charge Profile



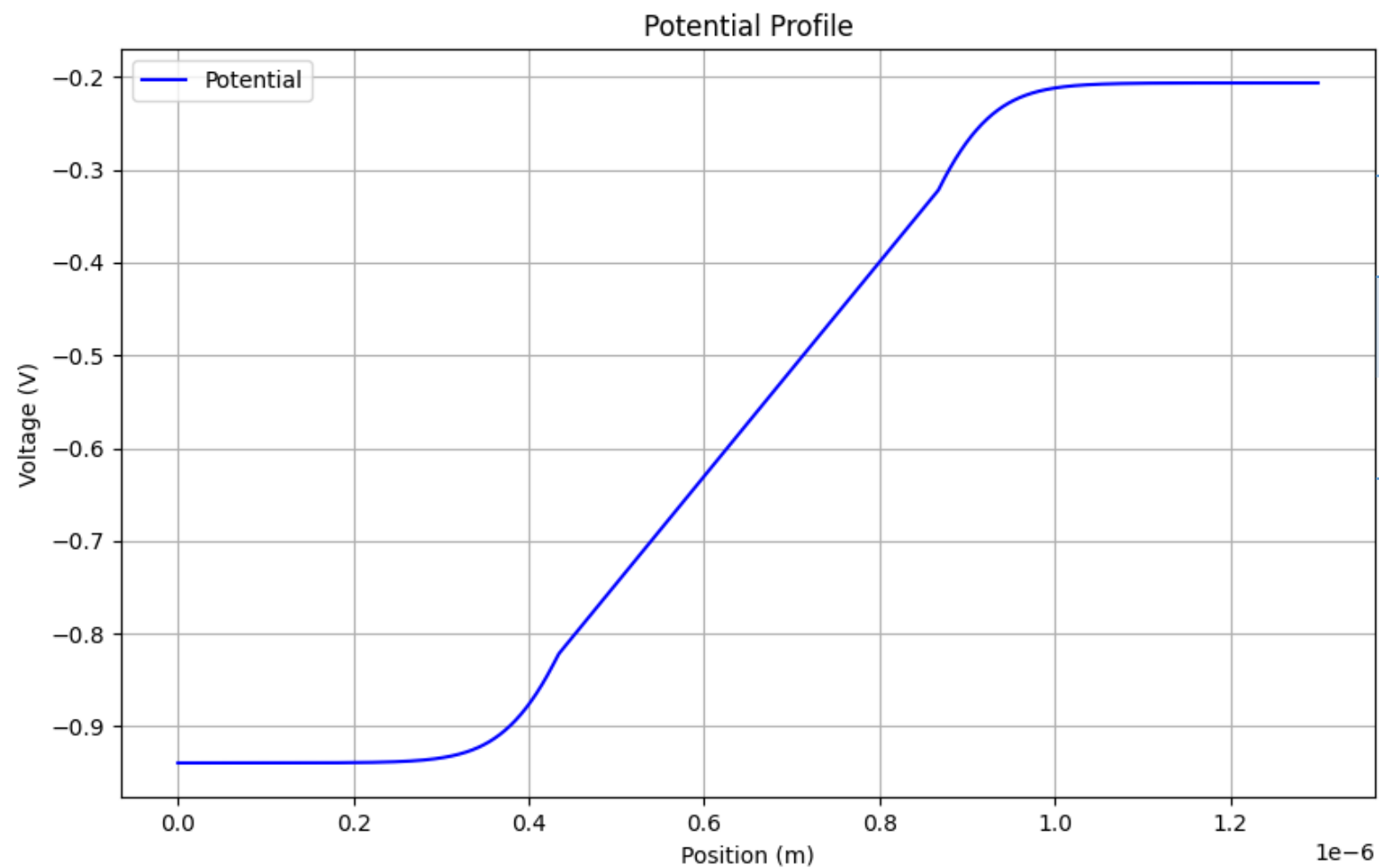
Electric Field Profile



Electric Field (V/m)

Maximum	-59.6051
Minimum	-1.66873e+06

Potential Profile



Voltage (V)	
-------------	--

Maximum	-0.20637
---------	----------

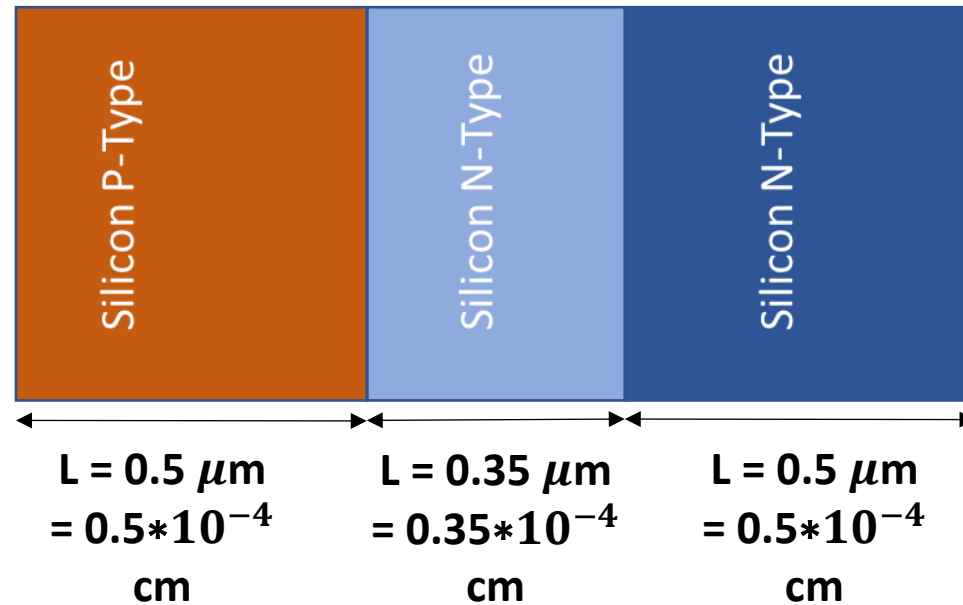
minimum	-0.93937
---------	----------

Potential Difference = -0.733 V

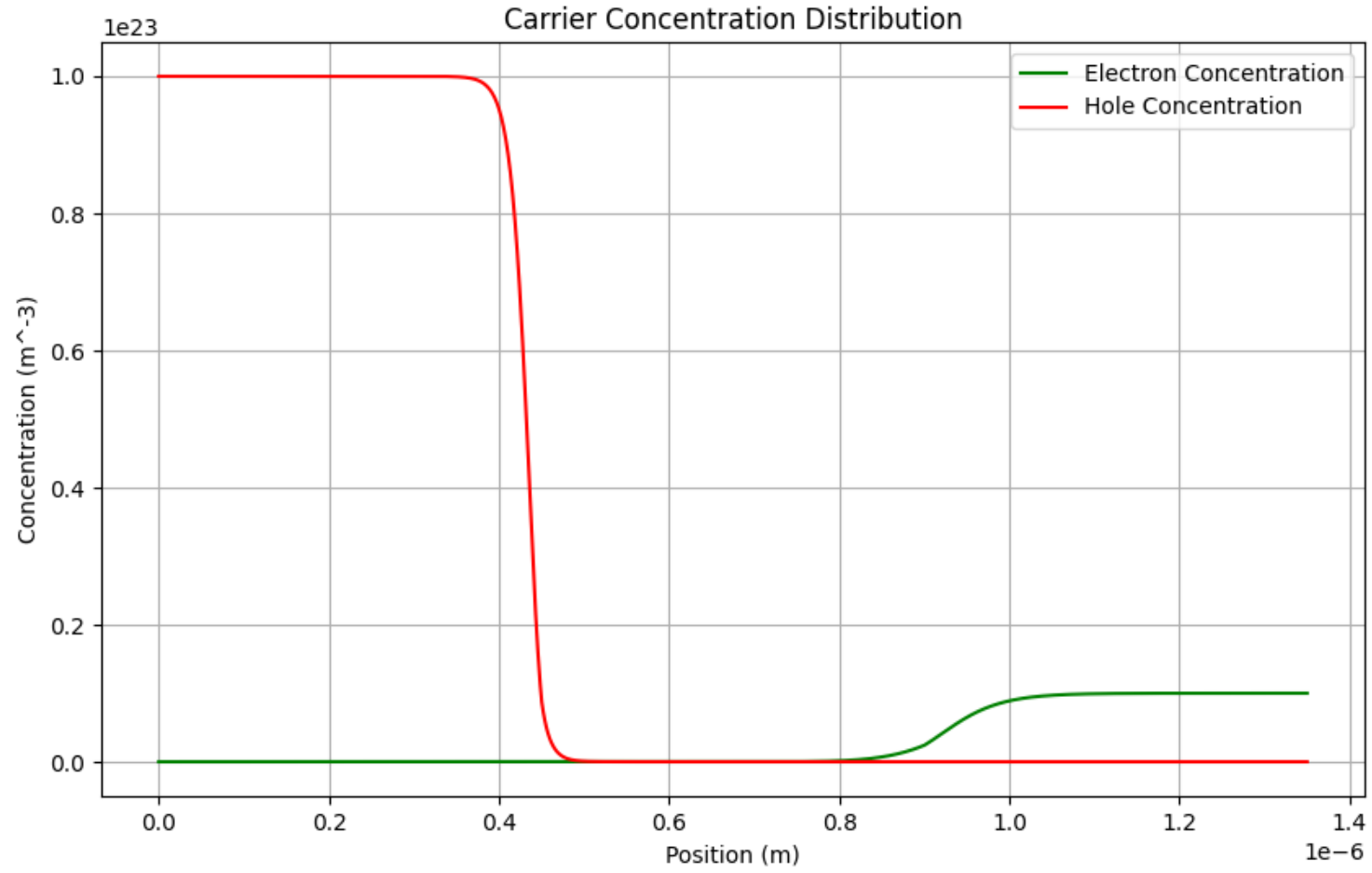


Doping profile of material-5

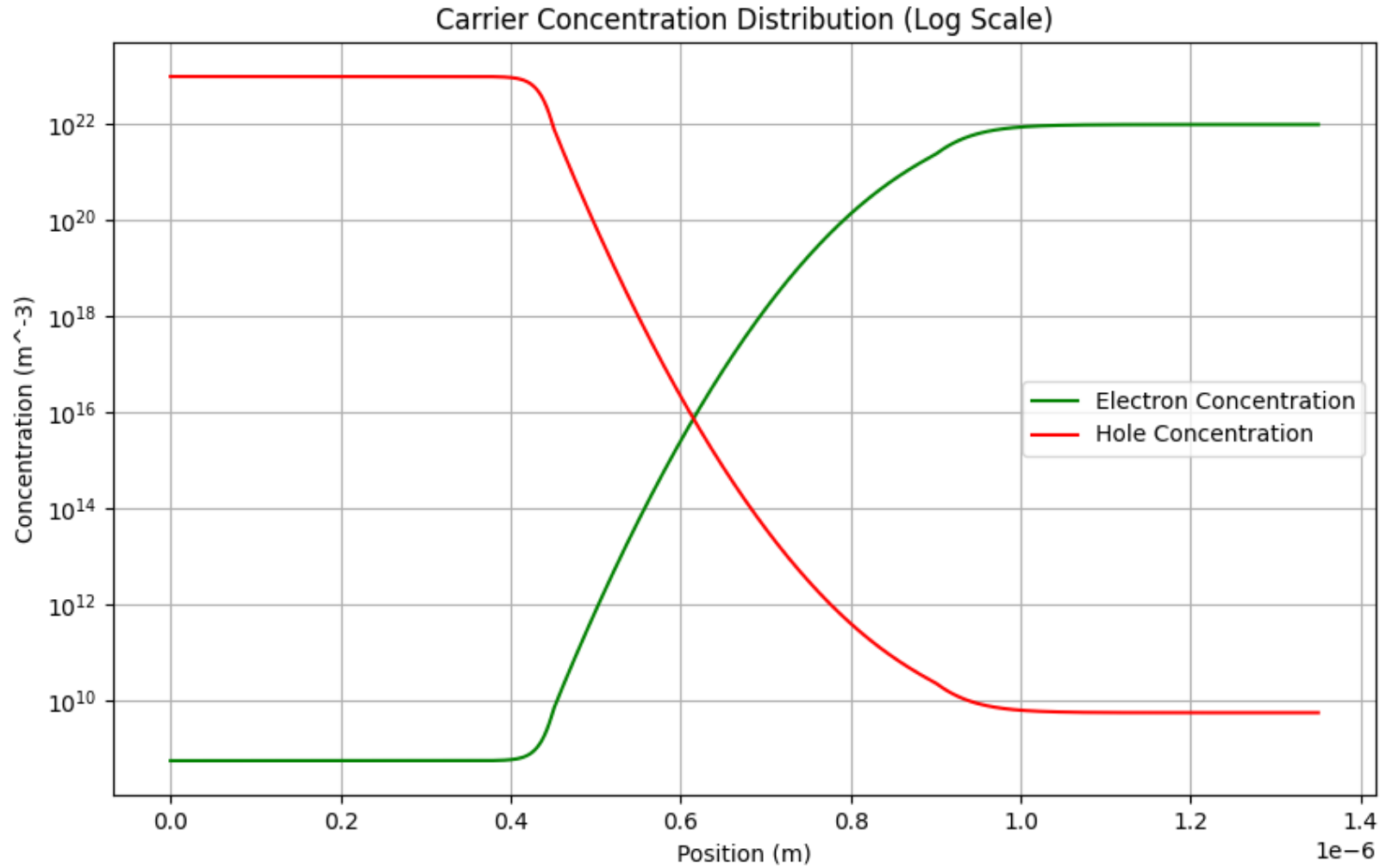
$$N_a = 10^{17} \text{ cm}^{-3} \quad N_d = 5 \cdot 10^{15} \text{ cm}^{-3} \quad N_d = 10^{16} \text{ cm}^{-3}$$



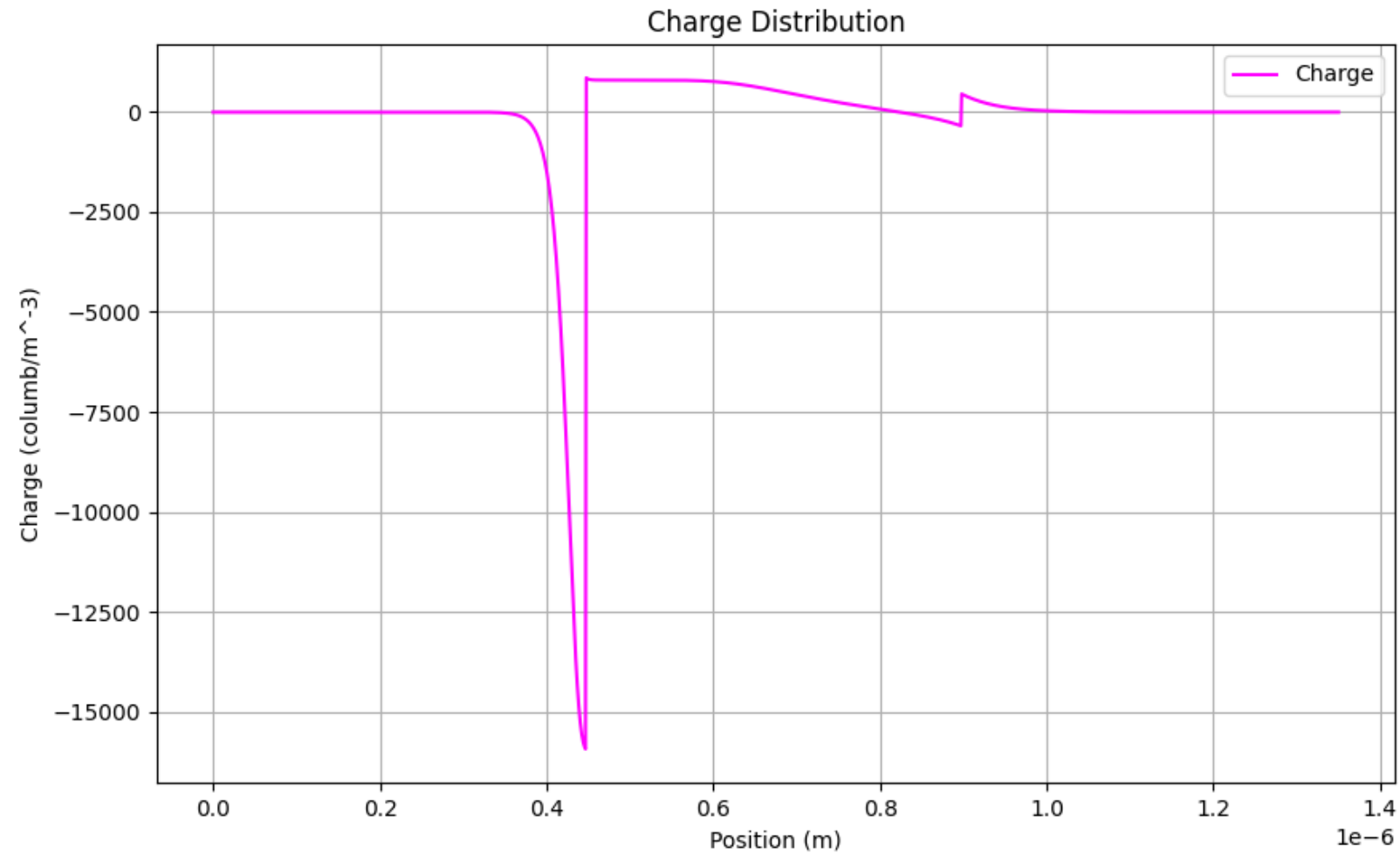
Electrons and holes Profile



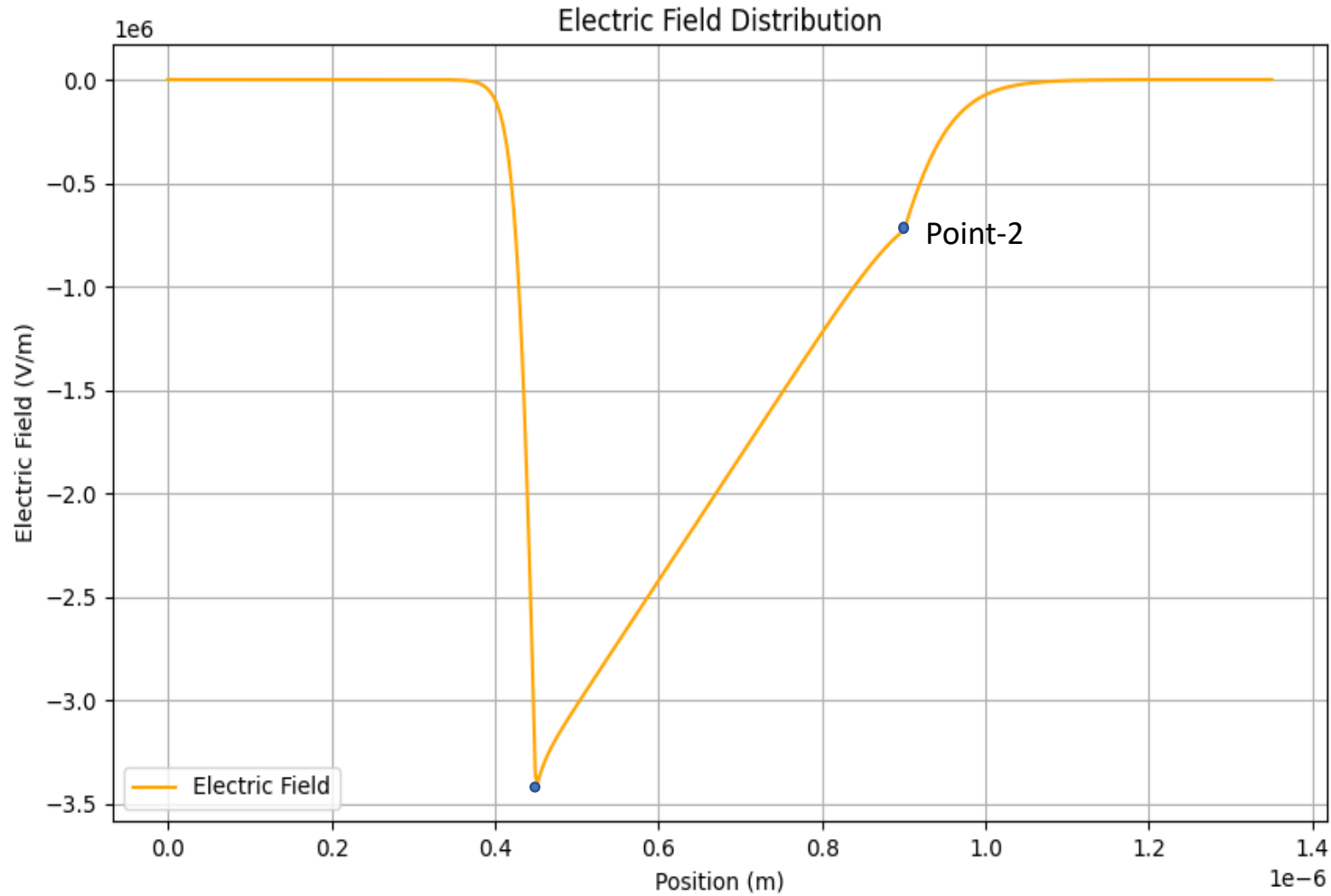
Electrons and Holes Profile



Charge Profile



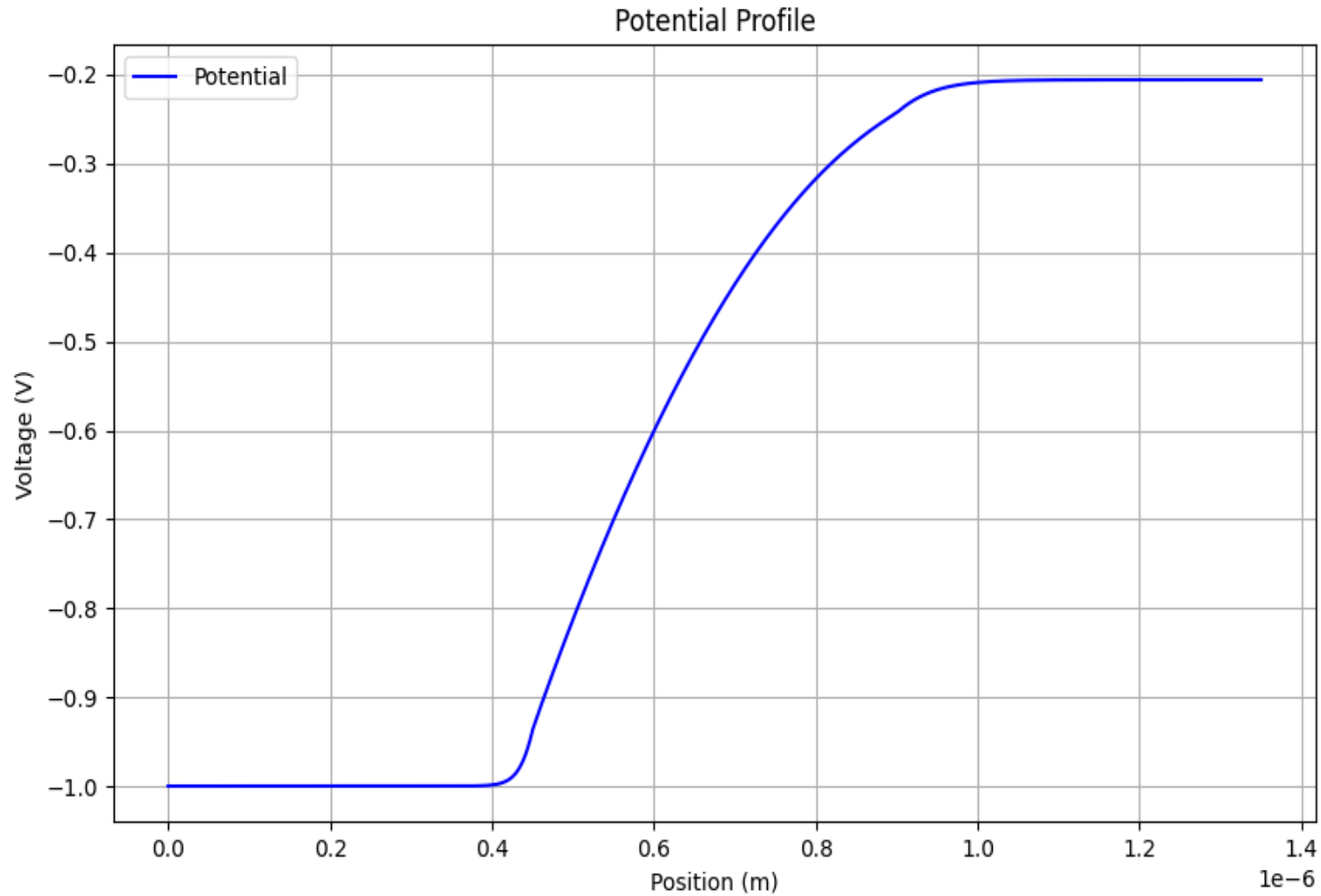
Electric Field Profile



Electric Field (V/m)	
Maximum	-1.88738e-06
Point-2	-7.13114e+5
Minimum	-3.41353e+6



Potential Profile



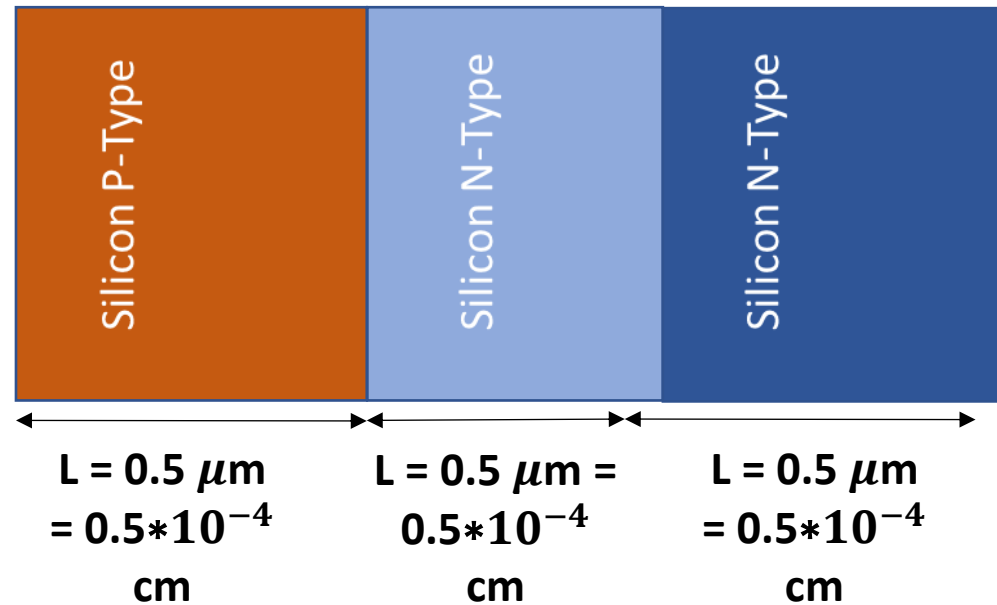
Voltage (V)	
Maximum	-0.206371
Minimum	-0.999245

Potential Difference = -0.7928 V

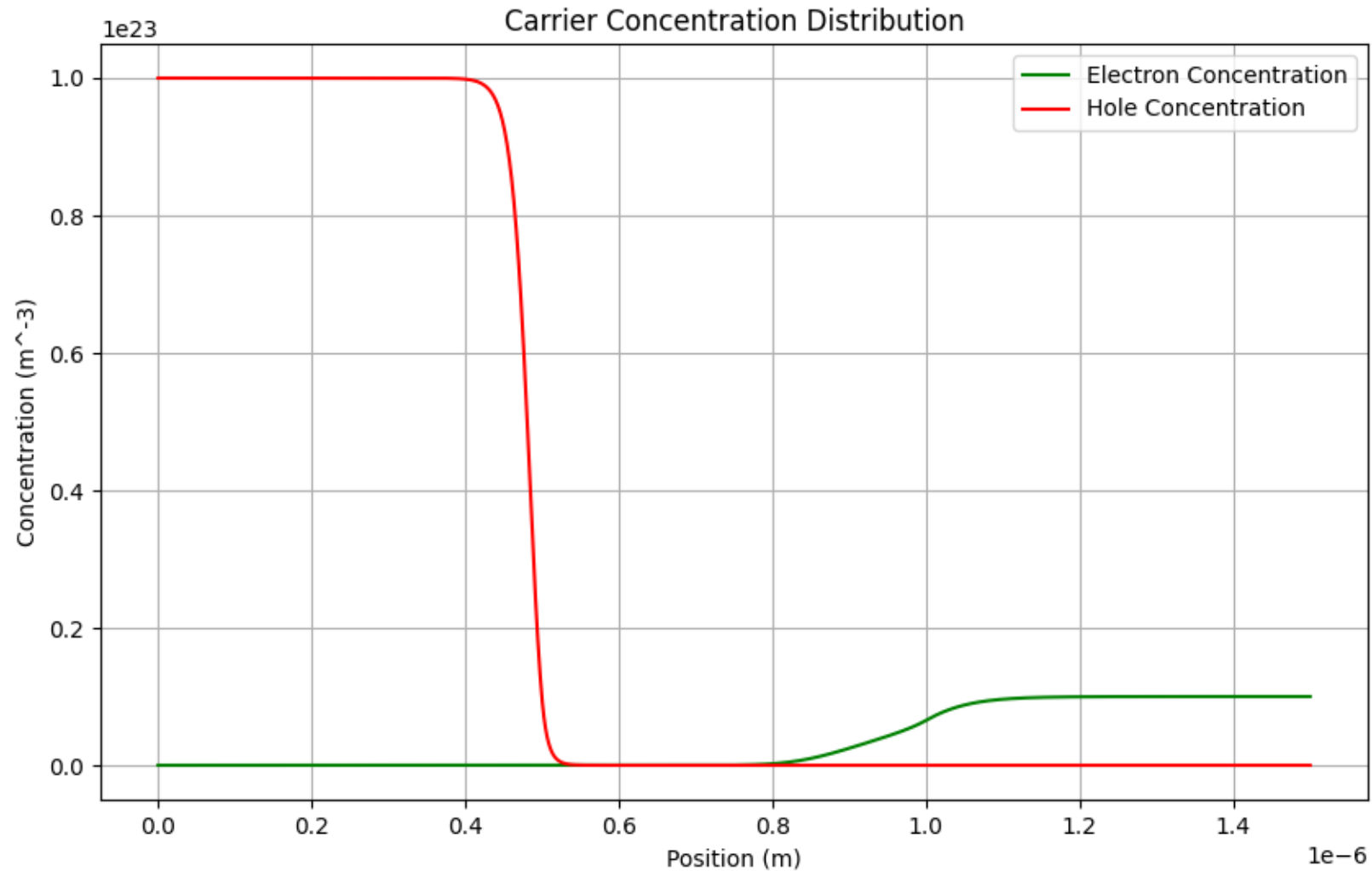


Doping profile of material-6

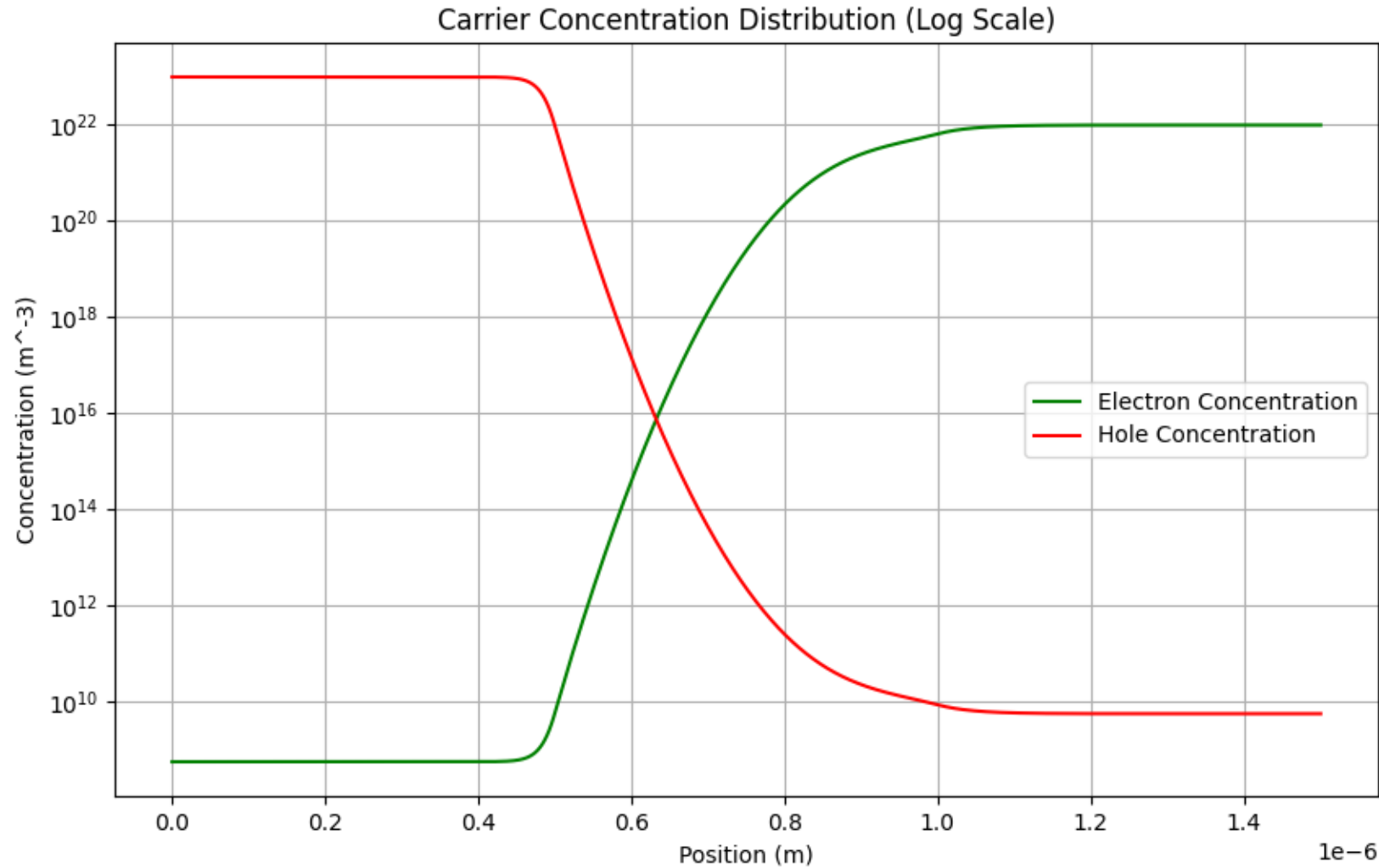
$$N_a = 10^{17} \text{ cm}^{-3} \quad N_d = 5 \cdot 10^{15} \text{ cm}^{-3} \quad N_d = 10^{16} \text{ cm}^{-3}$$



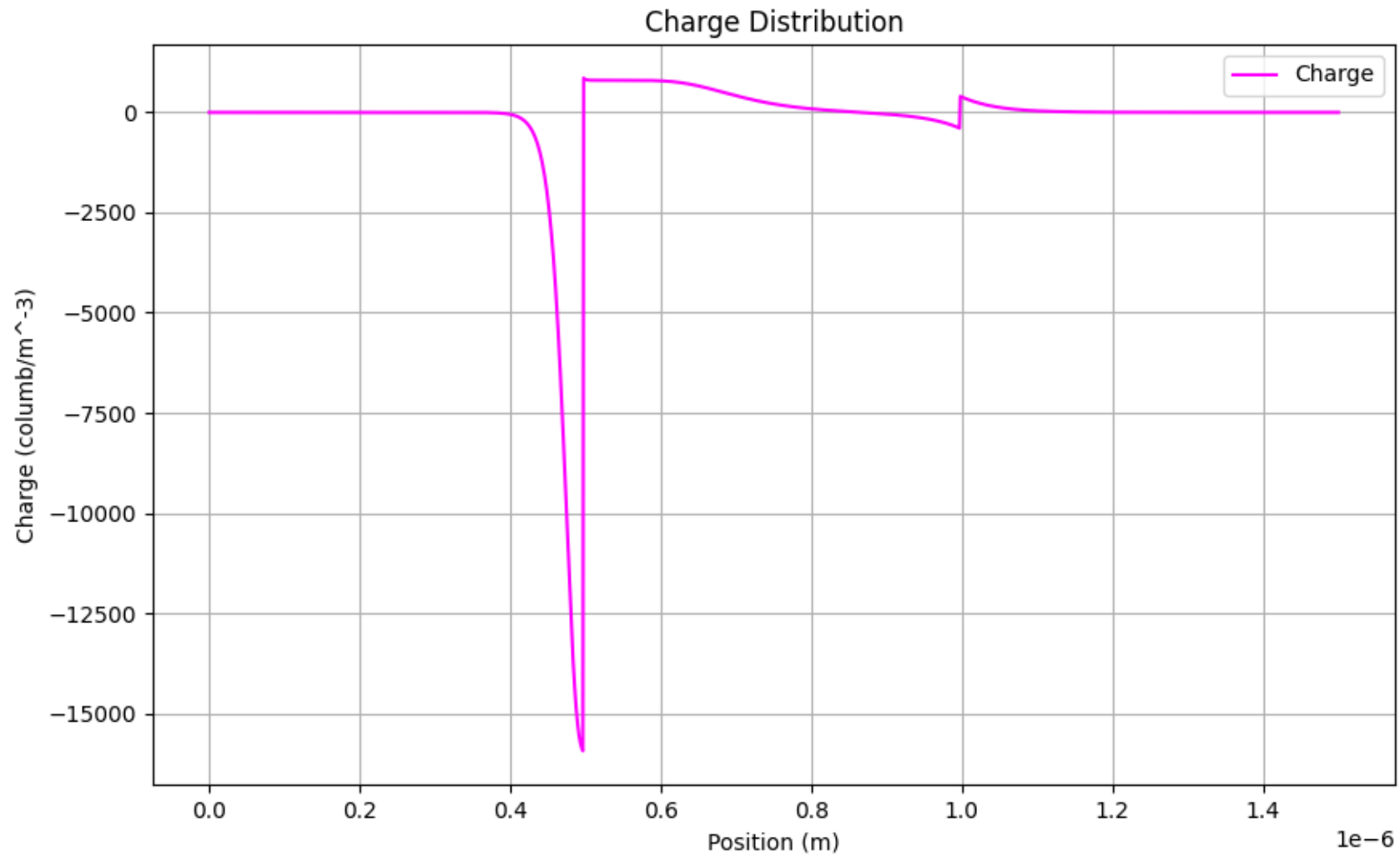
Electrons and holes Profile



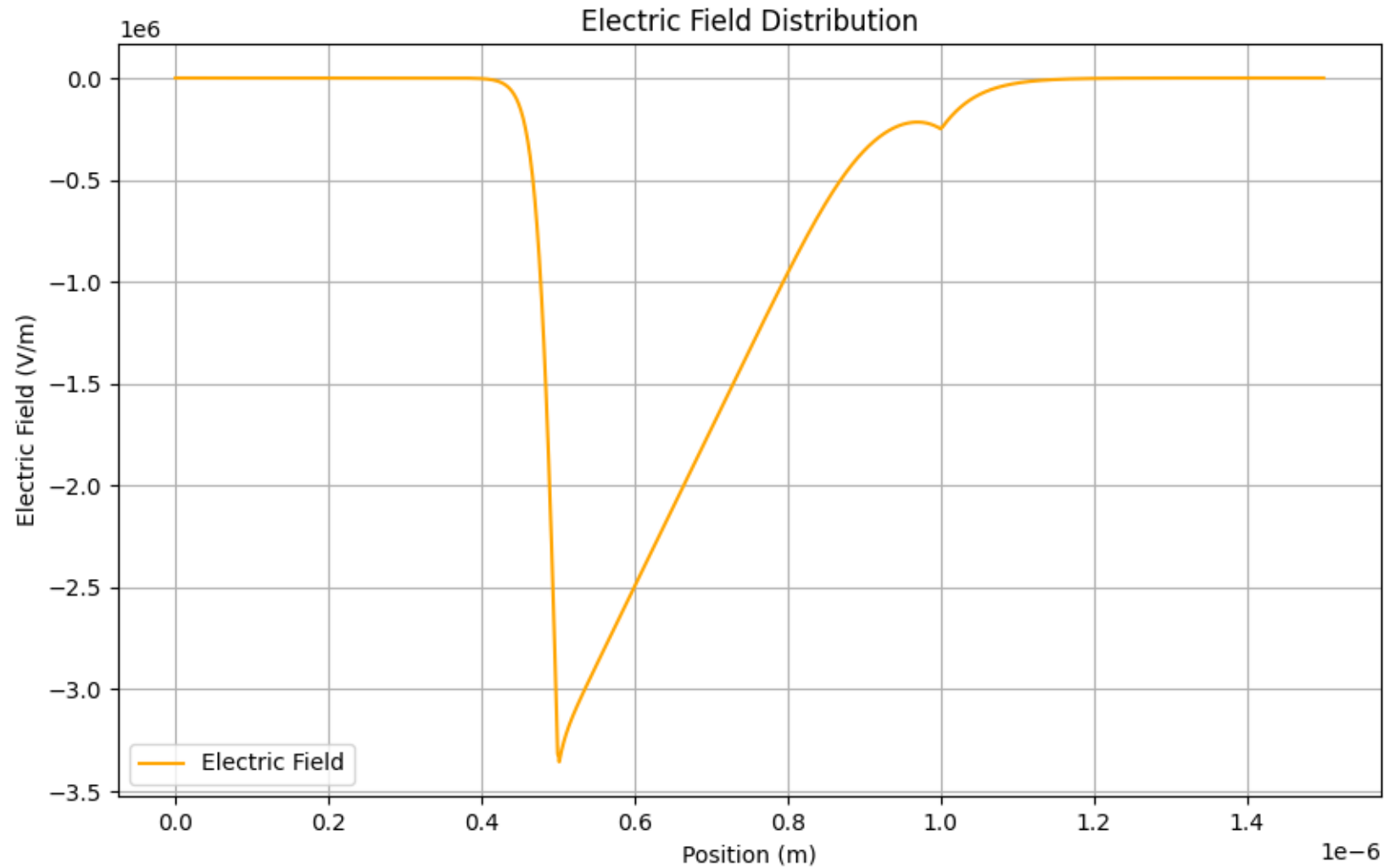
Electrons and holes Profile



Charge Profile



Electric Field Profile



Potential Profile

